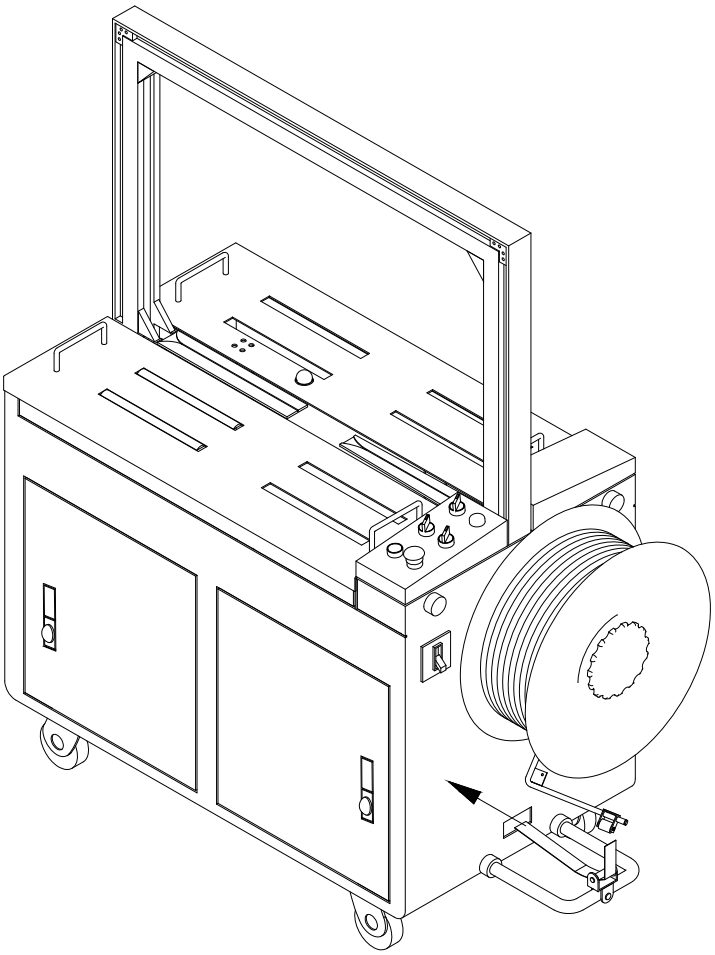
MH-101A Automatic Strapping Machine

Use, Maintenance and Spare Parts Handbook



Foreword

Thanks for purchasing and using our company's products. In order to ensure safety and correct usage of our products, please read the user manual carefully before operating.

This user manual is applicable to MH-101A strapping machine.

This user manual contains main technical parameters of the products, safety cautions, operation methods, maintenance and the methods to deal with some common faults. We have tried our best to ensure the comprehensiveness and accuracy of this manual when we prepare it. If you have any question or find insufficient or incorrect contents during the use, welcome to contact our company or our authorized agents directly. If a user has a different understanding, it shall be subject to the explanations of the technical department of our company.

The content of the user manual or part thereof may change without prior notice.

Please keep the user manual properly, for your reference to it at any time and to ensure the correct usage of this machine.

Without the written permission of our company, the content of the manual shall not be copied or changed, otherwise it will be considered as infringement.



The machine can only be operated by relevant personnel.

Daily safety instructions:

Please refer to the following information if you need to operate or maintain the machine.

Before operation

- → Read the manual and get prepared.
- → Check all the parts of the machine. If any broken, replace it with a new one (an original spare part is recommended).
- → Make sure that no tools or other items are on the machine and keep it clean.
- → Read operation and maintenance instructions carefully.
- → Read the explanatory phases and important identification about machine application carefully.

During operation

- → Do not wear loose-fitting clothes in case that they are entangled into the machine.
- → Do not carry out maintenance operation barefoot.
- → Do not put hands or other objects inside the working area.
- → Do not put anything else irrelevant to strapping on the machine.
- → Do not use the machine to strap products beyond the scope of application.

After operation

- → Make sure all of the switches are turned off.



The machine can only be operated by relevant personnel.

Service and safety regulations

- → Please keep the service contents.
- → Do not conduct settings or adding lubricants while the machine is in operation.
- → Please cut off the power supply before opening the cover board.
- → Do not install any additional parts unauthorized by our company for the sake of the normal operation of the machine.
- → Please use the original spare parts.
- → Please use a power supply system meeting the requirements of the machine.
- → Do not tear off or cover type specifications or safety warning and instruction signs on the machine.
- → Operate the machine according to the manual if necessary.



The machine can only be operated by relevant personnel.

ATTENTION!

For the normal running of the machine, please keep both the machine and its parts clean, regularly lubricate any of the parts that need lubrication, and make inspection and repair for electrical equipment.

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PART 1 MACHINE PROPERTY AND SAFETY INFORMATION

1.1 Performance parameter

Item	Parameters
	MH-101A
Power supply	AC380V/50Hz, 1000W/10A
Packing speed	≤2.5 second/beat
Tension	0-60kg (adjustable)
Bearing capacity*	80kg
Strap requirements	PP
	Width (9mm-15mm)±1mm, thickness (0.55mm-1.0mm) ±0.1mm
	Strap reel: width160-180mm, inner dia.:200mm-210mm, outer dia.:400mm-500mm
	Stretching resistance: Max.150kg
	Strap length/reel: 4000- 2000m
Packing specifications	Min. package size: W 80 mm × H 100 mm, max. size: within the size of the arch
Overall dimensions**	L1400mm×W628mm×H1418mm
Worktable height	750mm
Machine weight**	220kg
Working noise	⩽75dB(A)
Environmental conditions	Relative humidity≤90%, temperature 0-40℃
Adhesion	Bottom adhesion, hot melt method, adhesive surface≥90%, adhesive width≥20%, deviation of adhesive position≤2mm
Notes: ** standard arch size: W800mm×H600mm * subject to changes without further notice	

1.2 Nameplate

For feedback, please inform the manufacturer of the product order number and model, see the machine nameplate for details (Figure 1-1).

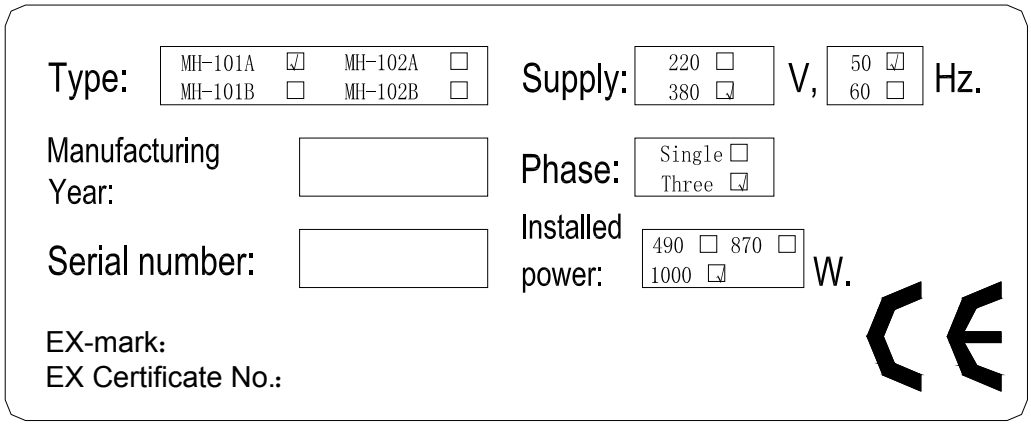


Figure 1-1

1.3 Overview and applications

The strapping machine is used to strap products using a polypropylene (PP) strapping (for machines). The products to be strapped are limited within the arch inner specifications (standard specifications: W800mm×H600mm). Be careful for rigidity of working materials, and don't try to strap wet, ashy, powder or too-heavy (usually max. load 80kg) packages.

This machine was designed for strapping only. We strongly recommend that this machine not be modified and/or used for any application other than for which it was designed. Pay attention to the load-bearing capacity described in the manual. Do not use the machine until you have known all items.

1.4 Safety considerations on machine operation

① Please check the power supply, and do not insert into a wrong power supply. The machine adopts a three-phase system, and the dual-color line is the grounding neutral wire for leakage protection.
② Do not touch the heater plate with hands directly.
③ Do not use water to wash the machine. If the workplace is wet, operating personnel should not work barefoot.
④ Do not change, increase or decrease the parts of the machine at will.
⑤ The strap conveying roller surface shall be free of oil.
⑥ Take out the power plug from the socket when the machine is not in use.
⑦ Main parts and components listed in the manual should be frequently lubricated with oil (see Clause 6.2)
⑧ The whole metal framework of the product shall be grounded reliably.
⑨ If it is used in the explosion-proof industry, it shall be equipped with explosion-proof motors and other electrical configuration.
⑩ It is not allowed to change or damage the structure and configuration of explosion-proof components, and the original equipment manufacturer shall be contacted for the maintenance of flameproof joints.
⑪ The use of the product shall comply with the product manual, the manuals of explosion-proof components, and the relevant provisions of Chinese standards GB 3836.13-1997, GB 3836.16-2006 and GB 50257-1996.

1.5 Machine radiation safety

Noise: ≤75 dB(A)

1.6 Safety signs

1.6.1 Safety sign labeling positions

Various prohibitions, warnings, instructions and tips are labeled on the machine, which remind people of paying attention to potential hazards or provide instructions, playing the role of avoiding possible dangers. (See Figure 1-2)

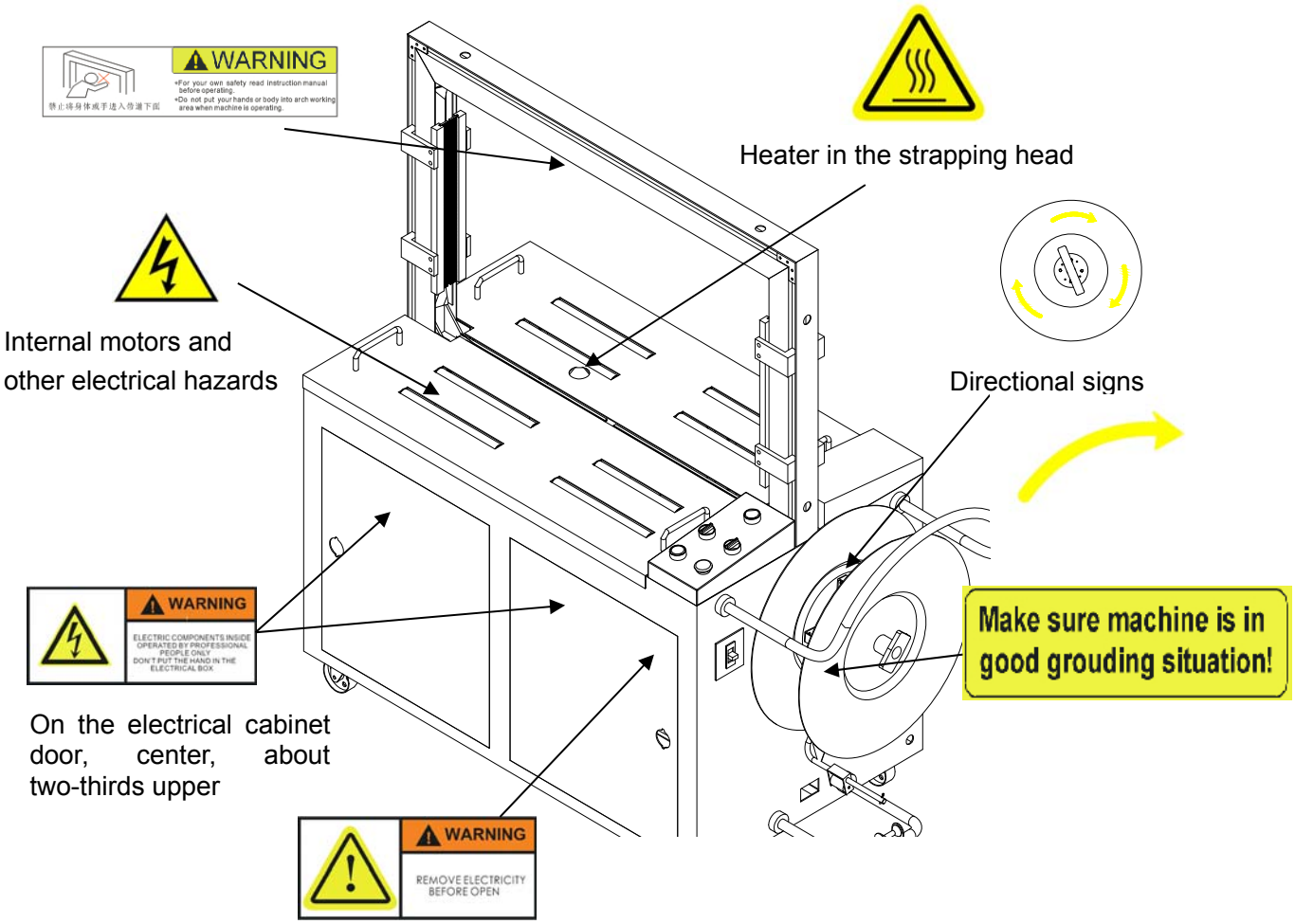


Figure 1-2

1.6.2 Safety signs description

Safety Signs Description

No.	Graphic Sign	Meaning	Remarks (labeling position)
1		**Warning Paying Attention to Safety**	In the parts where personal injuries may be easily caused
2		**Warning Electric Shock**	On the electrical equipment and lines where shock hazards may occur
3		**Caution Hot Surface**	On the parts with high temperature, like heating or heat generating parts
4		**Grounding**	The machine must be reliably grounded when it is used.
5		**Warning Be Careful**	Body or hands into the arch is prohibited.

PART 2 MACHINE SPECIFICATIONS, INSTALLATION, TRANSPORT AND STORAGE CONDITIONS

2.1 Specifications

2.1.1 **Model:** MH-101A

2.1.2 **Date of manufacture:** See the Certificate of Conformity

2.2 Installation

2.2.1 Installation conditions

The machine must be installed near a power supply and on a flat dry hard floor, fixed securely. The power supply must comply with what described on its nameplate.

The installation site shall meet the following conditions:

- Ambient temperature: 0℃-40℃
- Relative humidity: ≤90%
- Base plate: Max. loading 350kg/m^2
- Space: Min. 2,000mm×1,500mm
- Avoiding exposing to abnormal vibration
- Don’t use the machine in dangerous environment.
- Being far away from smoke, fire and corrosive substances, and being dry and well-ventilated, not exposing it to rain

2.2.2 Installation steps

① Installation of the aluminum alloy arch

Install the aluminum alloy arch as shown in Figure 2-1 into the frame slot as shown in Figure 2-2, and then fix it and the frame slot with M8X20 hexagonal socket screws, as shown in Figure 2-3.

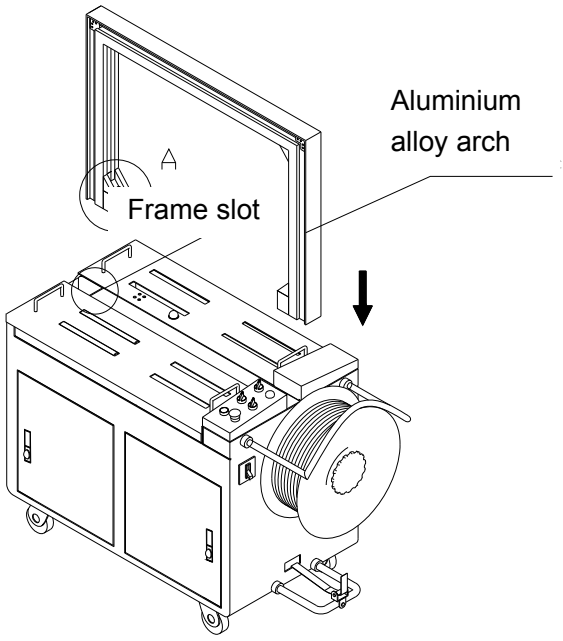


Figure 2-1

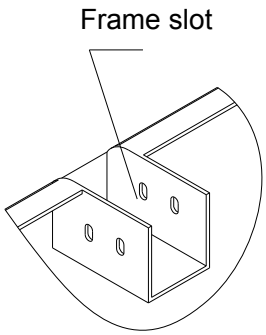


Figure 2-2

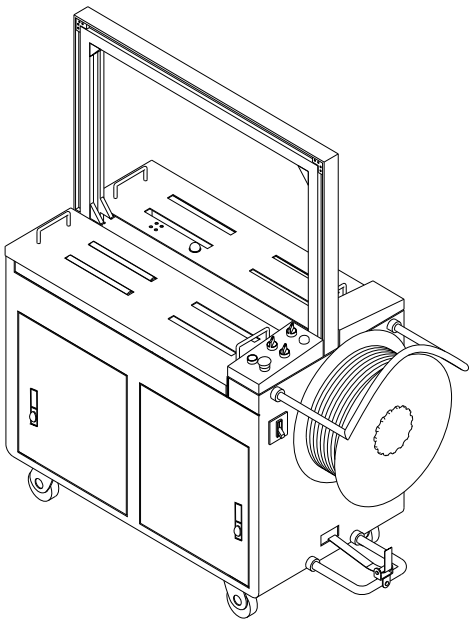


Figure 2-3

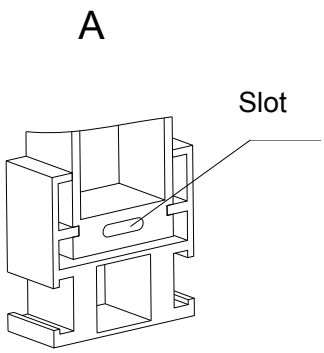


Figure 2-4

② **Installation of the left strap path**

Insert one end of the left strap path following the arrow direction shown in the figure into the aluminum alloy frame slot as shown in Figure 2-4, as shown in Figure 2-5. Then fix the other end onto the support plate of the strapping head as shown in Figure 2-6 with M5X12 hexagonal socket screws, as shown in Figure 2-7. Assure that the gap between the left strap path and T-shaped guide plate is H=2mm as shown in Figure 2-8.

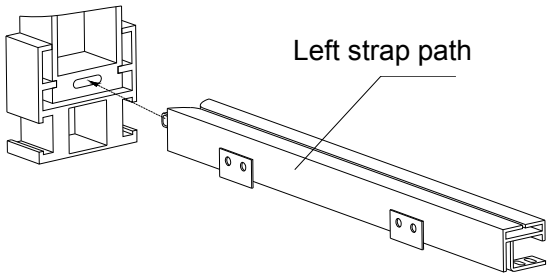


Figure 2-5

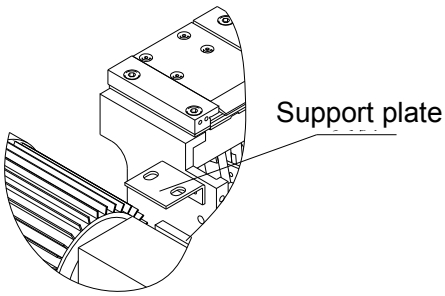


Figure 2-6

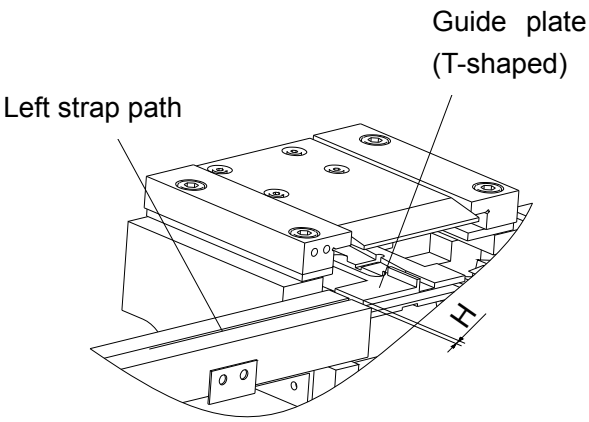


Figure 2-7

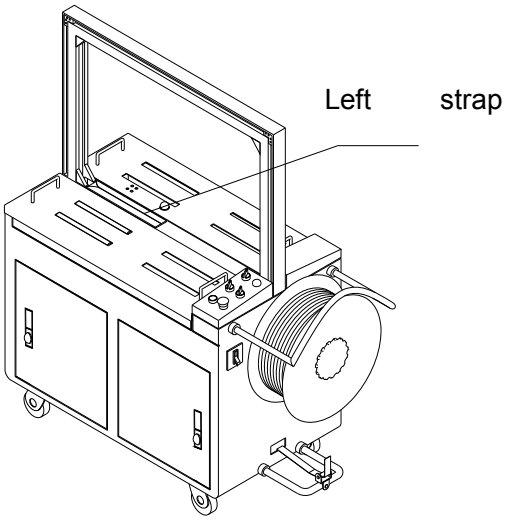
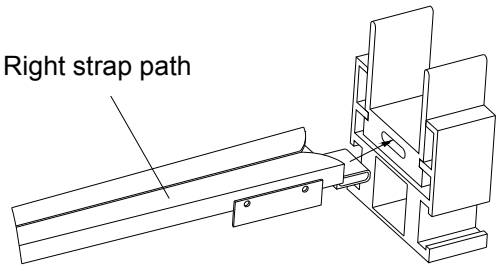


Figure 2-8

③ Installation of the right strap path

As shown in Figure 2-9, insert one end of the right strap path following the arrow direction into the aluminum alloy frame slot as shown in Figure 2-4, fixing them with screws, as shown in Figure 2-10.

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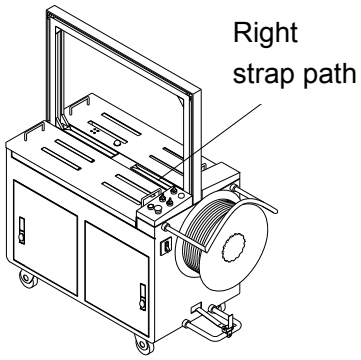
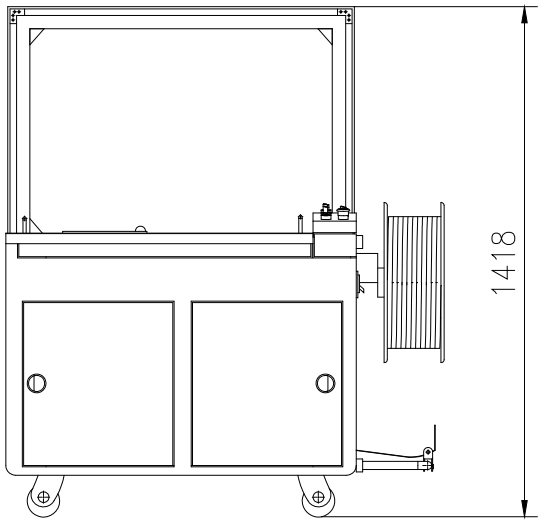
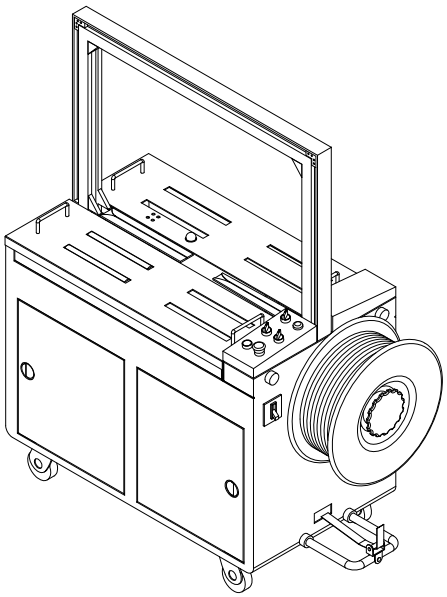


Figure 2-9 **Figure 2-10**

2.3 Transport and storage conditions

2.3.1 Machine diagram (see Figure 2-11)





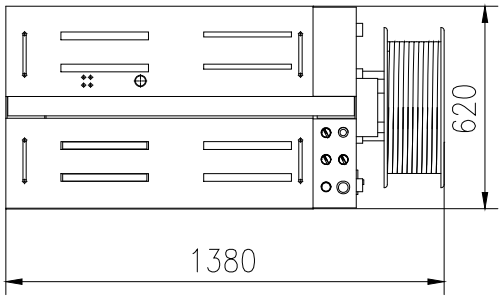


Figure 2-11

2.3.2 Transport

Transport it with a forklift truck (See Figure 2-12).

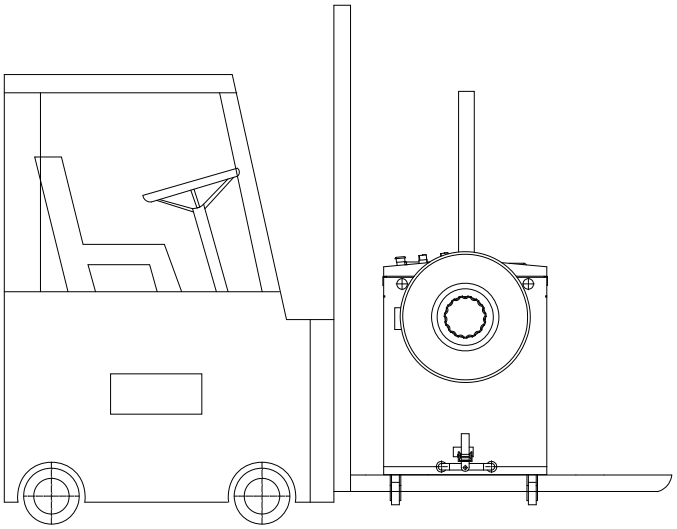


Figure 2-12

2.3.3 Storage conditions

For long-distance transportation or long-term storage of the machine, the machine must be packaged in a wooden or hard cardboard box, and fixed onto a solid pallet. Moisture absorbent shall be placed in the packaging box, and anti-rusting agent shall be sprayed inside the machine.

Do not place anything heavy on top of the wooden case or cardboard box. And the machine cannot be inverted during transportation and storage.

Place the machine on an even floor in order to avoid any distortion. The store environment must be dry. Do not expose the machine to extreme cold or heat environment.

After long-distance transportation or long-term storage of the machine, it is recommended that check all the moving parts manually before the machine is connected to the power supply, so as to avoid the overloading in electrical equipment.

PART 3 OPERATING THE MACHINE

3.1 Cautions before operation



Check if the power supply is in line with the one stated on the nameplate at the side of the machine. Check if the width of the PP strap used fits the inner width of strap track between the tables.



Check whether the direction of motor rotation is clockwise. You can identify the correct direction by the arc arrow marked on the strap feeding roller of the accumulator box. You must check the motor rotation direction if it becomes necessary to change the power socket or the wire plug.



While loading the PP strap reel onto the reel control unit, you must load it according to the direction indicated by the arrow marked on the outer flange. When the PP strap goes into the strap accumulator box and upper and lower rollers for strap feeding through the strap guide roller for strap feeding, the inner surface of the strap must be downward. Incorrect strap threading will result in machine operation failure.

1) It is not allowed to use the machine in the following circumstances such as:

①. When the cover board is not in its place;
②. When the machine is not stable.

2) It is not allowed to use the machine within the explosion scope of hazardous substances.

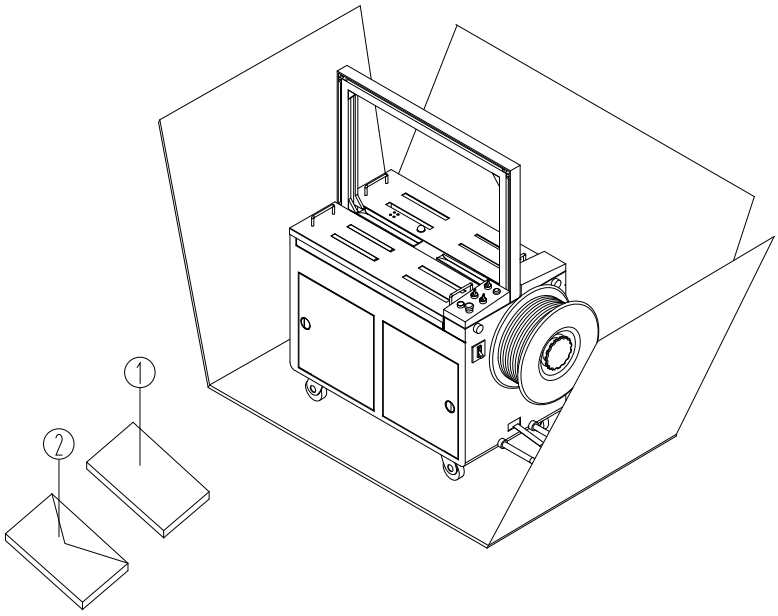
3) ATTENTION!

Do not wear bracelets, watches, finger rings, chain bracelets, neckties or laces while working, and workers shall be restricted from free movement to avoid injuries caused to themselves. The clothes of operators must comply with safety regulations when they are maintaining and operating the machine.

4) Pretreatment

After the machine is taken out from its package, as shown in the figure below, please check:

① There are not missing and damaged parts of the machine;
② Remove wrapping materials in the packaging box;
③ Check the model;
④ Check the accessories and the manual we provide with the machine; and
⑤ Report any damages caused during transportation.



①. One instruction manual
②. A toolkit

3.2 Check before operation

① Check fastening pieces are not loosened.
② Fill the decelerator with liquid lubricating oil and observe that the oil way is smooth (See Figure 3-1).

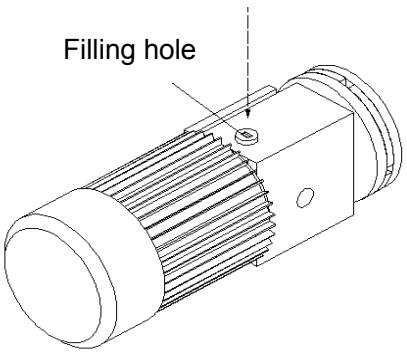


Figure 3-1

③ Check that motors and electrical equipment are dry with good insulation.

④. Check that the external power supply is fit for the power supply of the machine.

3.3 Idle running

After switching on the power, press the button “CONTINUOUS”, continuous idling running on, and check that proximity switches are in proper position.

① When the two proximity switches detect the strap withdrawal probe as shown in Figure 3-2, the main motor stops, the strap withdrawal motor starts, the machine starting to withdraw the strap.

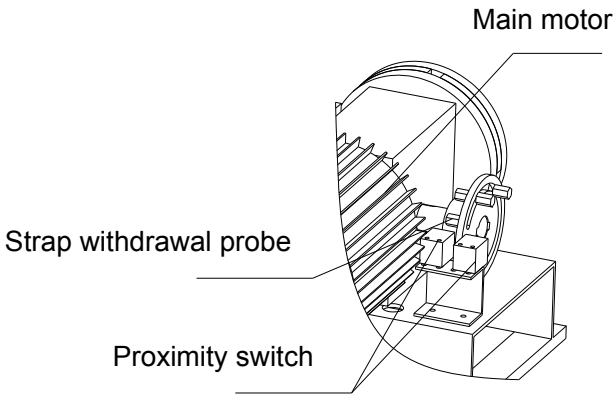


Figure 3-2

② When the two proximity switches detect the strap feed probe as shown in Figure 3-3, the machine starts to feed the strap.

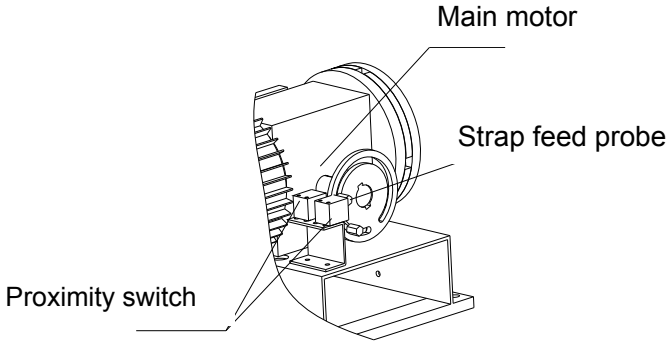


Figure 3-3

③ When the two proximity switches detect the stop probe as shown in Figure 3-4, the main motor stops.

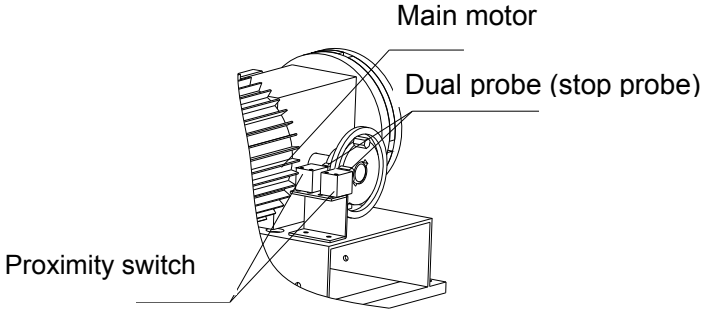


Figure 3-4

3.4 Operating steps

a. Thread the strap according to the method described in 3.6.
b. Switch on the power switch 1 as shown in Figure 3-5.
c. Put a package to be strapped on the machine, abutting against the stopper.
d. Apply the strap around the package, inserting it into the strap inlet, and then the machine carries out strapping automatically.
   Manually put a package to be strapped on the worktable, apply the strap around the package, inserting it into the strap inlet, and then the machine carries out strapping automatically.
   Attention: The package to be strapped shall be placed on the machine, abutting against the stopper.

3.5 Operating control unit

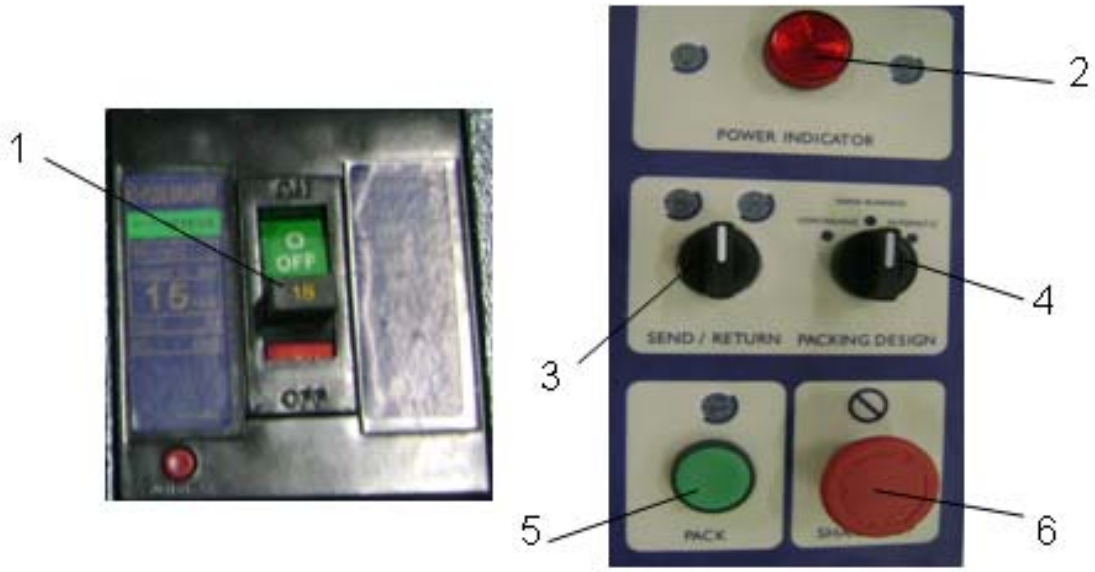


Figure 3-5

As shown in Figure 3-5:

① **Power switch**

② **Power indicator**
If the indicator light is on, it indicates the power supply is not turned off.

③ **Strap feeding and withdrawing switch**

④ **Operation mode selector switch (Continuous/Automatic/Manual)**
When the switch “PACKING DESIGN” is turned to “CONTINUOUS”, the machine carries out strapping continually at set intervals (adjustable); when to “AUTOMATIC”, it carries out strapping once the float ball switch is touched; when to ” HAND-RUNNING”, you need press the button “PACK” to carry out strapping.

⑤ **Pack button**

⑥ **Emergency stop button**
No matter what state the machine is in, if the machine has to be stopped

immediately, just press this button. To restart the machine, turn for an angle in the arrow direction shown on the button to deactivate the emergency stop first.

3.6 Strap loading and threading method



During PP strap reel installation, main power switch must be turned off. When the PP strap goes through the strap guide roller in the accumulator box, its internal surface must be downward.

3.6.1 PP strap reel loading

As shown in Figure 3-6:

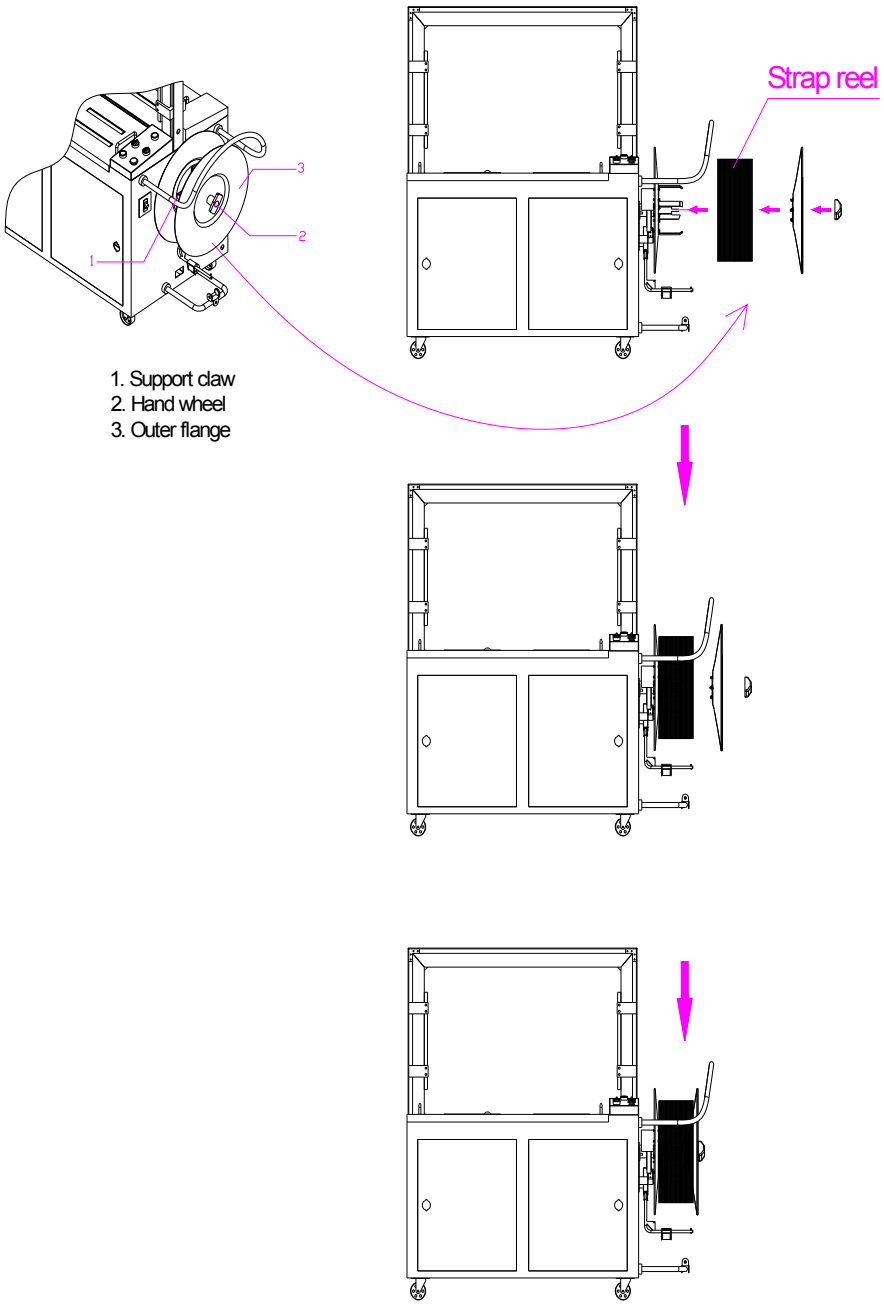


Figure 3-6

3.6.2 PP strap threading

① After loading a strap reel onto the strap control unit, as shown in Figure 3-6 and after the strap head enters the strap inlet, you have to turn the knob in the strap pre-feeding mechanism in the clockwise direction to make the two strap conveying rollers have a movement for a distance, so that the strap head can pass between the two rollers to go to the strap accumulator box. Pull the strap head by hand and then turn the emergency stop button on the control panel and the strap accumulator box will be filled with the strap automatically.

② After the strap head is pulled out from the upper opening of the strap accumulator box, insert it into the tension crank as shown in Figure 3-7, and let it pass between the two conveying rollers. At this time, turn the button 3 on the control panel to the position “SEND”, and the strap path will be full of the strap, thus strap feeding is finished, and the machine is ready for strapping.

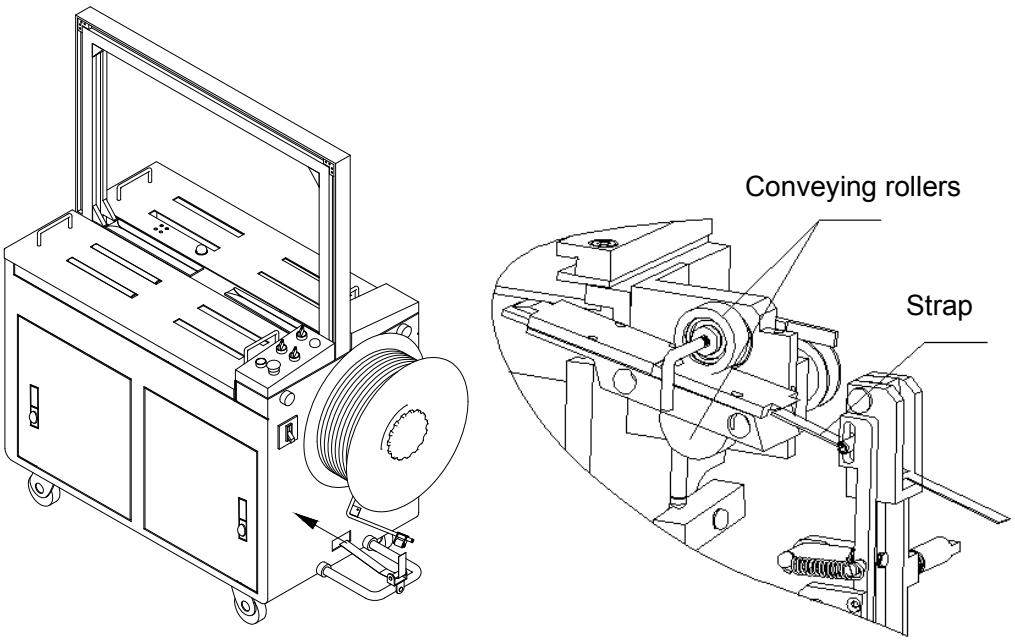


Figure 3-7

3.7 Working position illustration

When working , the operator shall stand in front of the machine, about 100mm away from it (at a distance where he is able to operate it freely), see Figure 3-8.

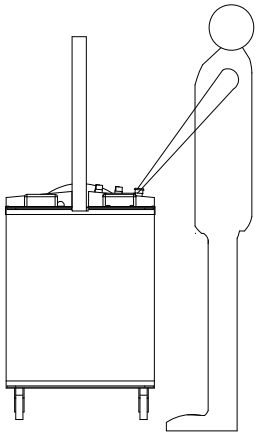


Figure 3-8

3.8 Connecting to power supply, motor device and putting into use

Attention:

Before connecting to power supply, please check that the power switch of the machine is turned off (Figure 3-9, Position 2) and there are not other objects on the worktable.

Before use:

1) Check that the motors accord with the power supply requirements of the machine.
2) Check that there are not loosened fasteners.
3) Check that the motor device and electrical equipment are dry with good insulation.
4) The power switch must be in a safe position (Figure 3-9, Position 2).

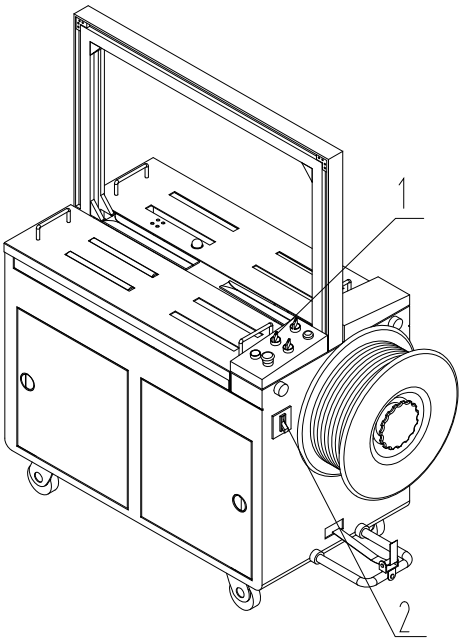


Figure 3-9

In operation:

1) Press the start button to start and use the machine.
2) Check the machine runs well 2-5 minutes after connecting to the power supply.
3) Press the manual strap feed button to see that the movement of the strap is smooth (Figure 3-9, Position 1).

CAUTION!
The machine can only use the plug supplied with it, so do not take down the plug on the cord.

PART 4 ADJUSTMENTS

4.1 PP strap tension force adjustment

Tension force adjustment: This position is located at the position after the front right door is opened (see Figure 4-1). Generally, the knob to adjust the tension force is locked and can't be turned. When adjusting the tension force, first forcibly push the knob to the bottom, and then you can freely turn the knob. The tension force is decreased as you turn the knob in the clockwise direction, and increased in anticlockwise direction. In general, on delivery, the position of the bearing and the positioning plate as shown in the figure below is set, so generally it is not necessary to adjust their relative position.

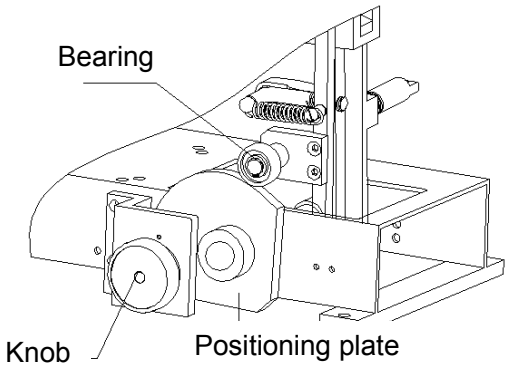


Figure 4-1

4.2 Heating temperature adjustment

Temperature adjustment: There is a potentiometer over the operating panel cover of the machine, which is used to adjust the tension strength of strapping, and you can see it after opening the cover plate. The temperature at 2 is the minimum, and at 10 the maximum. Turn it clockwise, and the temperature will rise. If the temperature is too high or too low, the adhesion effects will not be good, so do not turn it sharply unless the voltage gap is too big. Generally, it is set at between 4 and 6, as shown in Figure 4-2.

Attention: Do not turn it sharply.

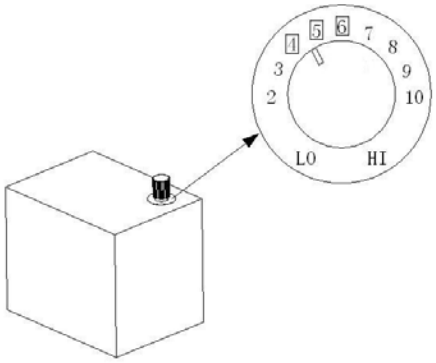


Figure 4-2

4.3 Electromagnetic brake gap adjustment

Electromagnetic brake gap adjustment: The gap between the brake pad and the brake shall be controlled at 0.3-0.5mm; if too large or too small, it will result in poor braking of the brake.

PART 5 TROUBLESHOOTING

5.1 Possible faults and troubleshooting

Possible faults under normal use conditions:

1) Adhesive effect of the strap is not so good:

- **a.** Temperature of the heater plate is too high or too low.
- **b.** Deformation of the heater plate: If the heater plate is deformed, it can't be inserted between the two layers of the strap, resulting the failure to bond the two layers of the strap (See Figure 5-1).

c. Poor adhesion due to strap not fed to the proper position: The strap head does not reach the preset position, so the adhesion effect is not good.

d. The strap used is too thin, so that the pressure is not enough. If this kind of strap is used, a spacer not thicker than 0.5mm should be added on the middle bar (See Figure 5-2).

e. The pressure by the middle bar is not large enough: The compression spring in the middle bar is fractured, decreasing the pressure.

f. Split strap head: The strap head is split because the tension force is too strong, so you have to properly adjust the tension force (See Figure 5-3).

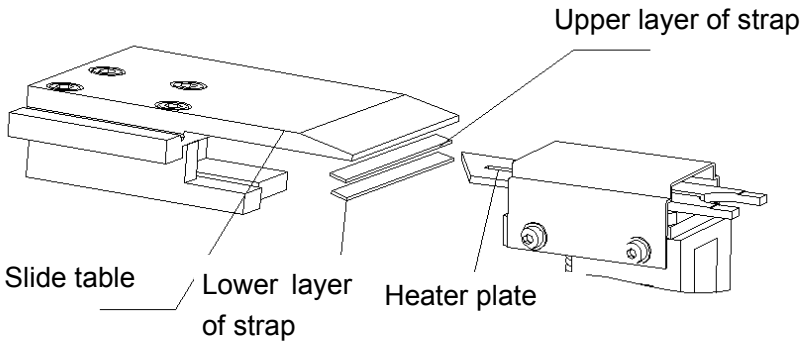
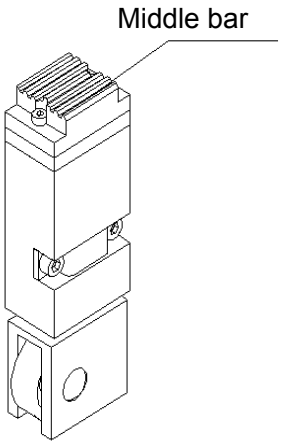


Figure 5-1



F**igure 5-2**

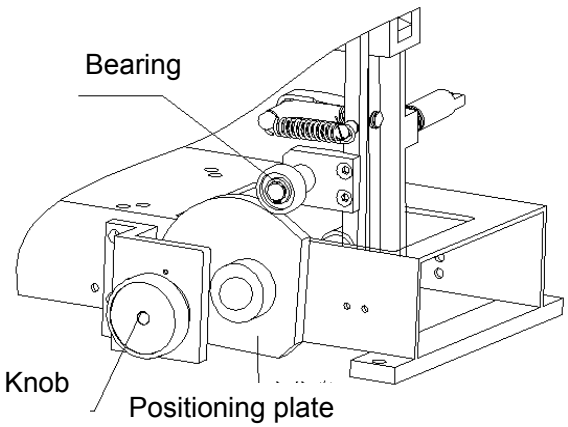


Figure 5-3

2) Strap is not fed to the proper position

a. Tension spring force of the pry bar is too strong or too weak: properly adjust the pressure of roller (see Figure 5-4)

Pry bar

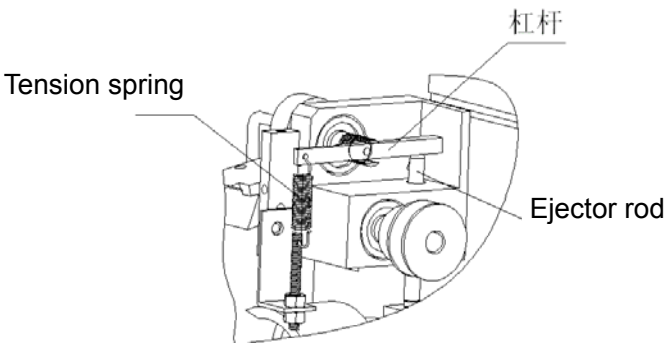


Figure 5-4

b. The position of ejector rod is too high: In the case of not withdrawing the strap, the gap between the ejector rod and the pry bar should be H=0.3mm (Figure 5-5). If due to improper adjustment, the ejector rod always holds against the pry bar and the holding force is very strong, it will cause the phenomenon that the strap is not fed to the proper position.

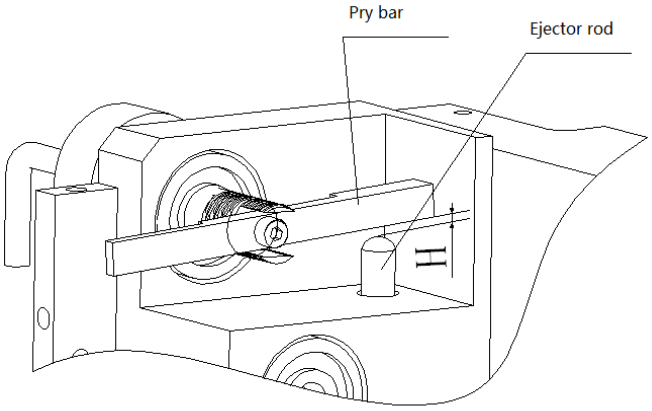


Figure 5-5

c. Short of strap storage volume in the strap accumulator box: If there is not enough strap, it will cause the case that the strap is not fed to the proper position.
Reasons: (1) improper adjustment of strap pre-feeding mechanism. Adjust strap storage volume properly.
(2) Fault of the strap pre-feeding mechanism or strap accumulator box problem. Find out the fault and solve it.

d. Split strap head: If the strap head is split, it will not run smoothly in the strap path and lead to the situation that the strap is not fed to the proper position. The reason is that the tension force is too strong. Therefore, properly adjusting the tension force will eliminate this situation (See Figure 5-3).

e. Strap-pressing plate is too low (See Figure 5-6): Strap-pressing plate is set to make strap feeding convenient. If it is adjusted too low, making the gap between it and the crank undersurface too small, which affects free feeding and withdrawing of the strap, it will cause this situation. At this time, properly raise the strap-pressing plate.

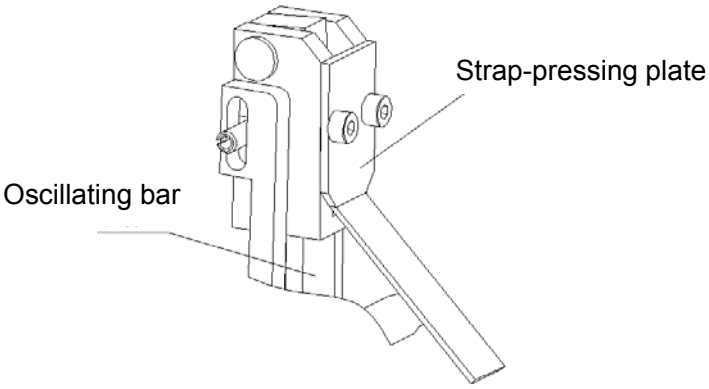


Figure 5-6

f. Middle bar is not fixed firmly: The middle bar is displaced as it is not fixed firmly, closely leaning against another bar, thus it cannot reset to the proper position and causes T-shaped guide plate's failure to reset, so that the strap head cannot go forward smoothly without the guide of the guide plate and leads to the situation that the strap is not fed to the proper position. Therefore, the middle bar should be fixed firmly after adjustment or reassembly (See Figure 5-2).

g. Quality problem of strap: Width, thickness, bending of strap may also cause this situation.
Judging method: Feed a strap manually, if it can't reach the proper position, the quality of the strap is not fit for the requirements of the machine; If it goes to the position smoothly, check for other troubles.

h. Strap can't be tensioned properly (See Figure 6-3 & Figure 5-3):
① Tension adjustment device is not adjusted for a proper tension;
② Strap clamp block is worn seriously;
③ Teeth groove of the strap clamp block is full of strap scraps;
④ Tension spring of pull bar is fractured;
⑤ Driving bearing is damaged;
⑥ Torsion spring is fractured.

i. Strap goes loosely:
① Strap withdrawing time is not enough;
② Strap withdrawing force is too strong or too weak. Check the gap H between the strap withdrawing ejector rod and the pry bar (See Figure 5-5);
③ Resistance of arch is too large;
④ Motor belt is too loose.

3) Power light is not on:
① Check that the power plug contacts well with the power socket;
② Check that the air switch is switched on.

4) The machine can't carry out strapping:
Check that the emergency stop button is not pressed down.

5.2 Malfunction warning

ATTENTION!

If the machine breaks down, please ask professional personnel who must have professional knowledge in both electrical and mechanical fields for troubleshooting. Make sure there is enough room around the machine before maintenance so that the maintenance personnel can avoid damage when maintaining.

Follow the following steps when conducting maintenance:

1) Stop the machine.
2) Cut off power supply.
3) Take down other mechanical parts of the machine.
4) Take out the machine from equipment.
5) Fasten moving parts, and prepare to transport the machine.

If the machine is out of service, please follow the procedures below:

1) Take down protective tools.
2) Classify the parts by material (metal, plastic, rubber, etc.), then make garbage classification.

5.3 Standard accessories supplied with the machine

- Socket head wrench: 1 SET
- Straight screwdriver: 1
- Cross screwdriver: 1
- Heater tension spring 1
- Brake disc spring: 1
- Slide table tension spring: 1
- Fuse (1A) : 1
- Fuse (2A) : 1
- Fuse (3A) : 1
- Piezoresistor: 1
- Diode: 1
- Strap path centering screw: 2
- Cover set screw: 1
- Sliding block snap spring: 2
- Strap pre-feeding snap spring: 1
- Middle bar compression spring: 1 big, 1 small
- Instruction manual: 1

PART 6 MAINTENANCE

6.1 Daily maintenance

Regular and correct maintenance and care to the strapping machine not only extends its service life but also reduces the breakdown of the machine, thus the working efficiency is improved.

Strapping machines work very fast, and each machine has to work for 16 hours every day when there is a large amount of work, thus their parts should often be in a good lubrication state. The following parts need to be lubricated once every week, with oil brand No. N68. The other parts not listed in this manual also need to be lubricated at irregular intervals.

6.2 Lubrication

Before lubricating, clean the parts needing lubrication with cloth material like a towel.

a. Slide table

- Clean the grooves between the slide table and the left and right guide rails.
- Apply a little stitching machine oil or lighter lubricating oil in the gap between the slide table and the left and right guide rails.
- If the slide table can move smoothly but fails to return to its home position, check that the tension spring is not exhausted, if it is, replace it or cut it short.

b. Guide plate (Figure 6-1)

Take out the guide plate, and lubricate its four sides playing the role of guide with lubricating oil, and then put it into the guide rails.

c. Running surface of cams and their rollers

d. Heater guide rail (Figure 6-2)

e. Strap clamp block (Figure 6-3)

Oil the two sides and the bore of clamp block.

f. Parts with shafts, spindles, and bearings shall be lubricated once every three months

g. Grease in the reducer gearbox should be supplied once a year.

h. Attention: The following parts shall be free of oil:

. ① Strap feed and withdrawal roller ② Electromagnetic brake

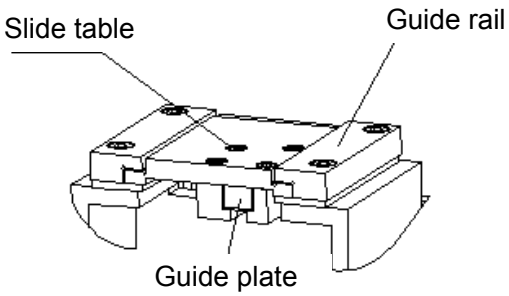


Figure 6-1

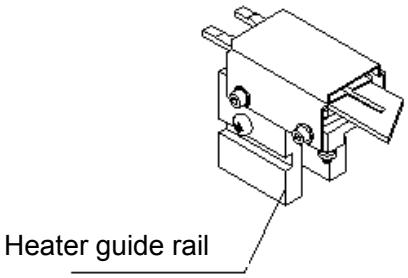


Figure 6-2

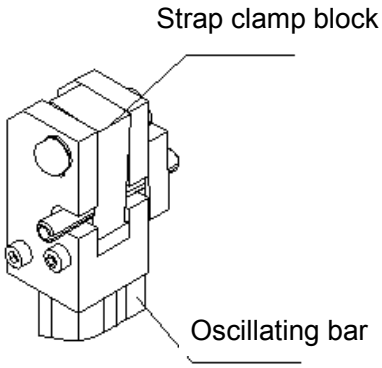


Figure 6-3

i. Bar guide assembly

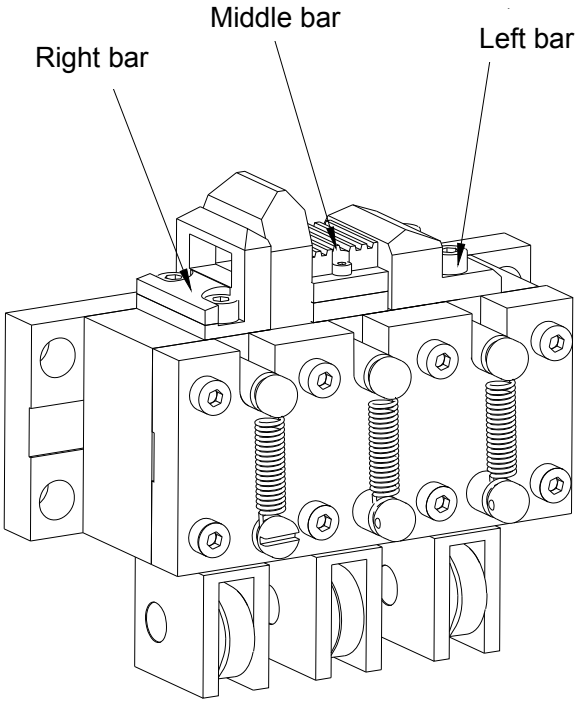
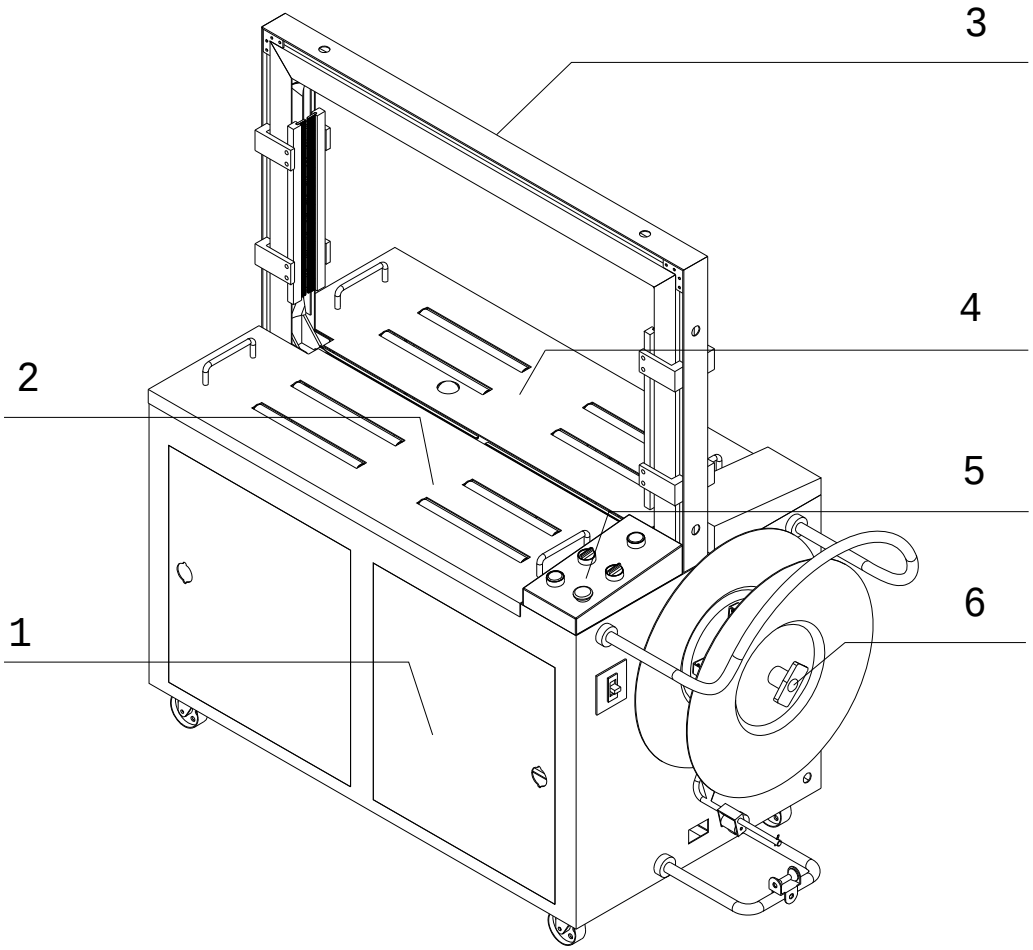


Figure 6-4

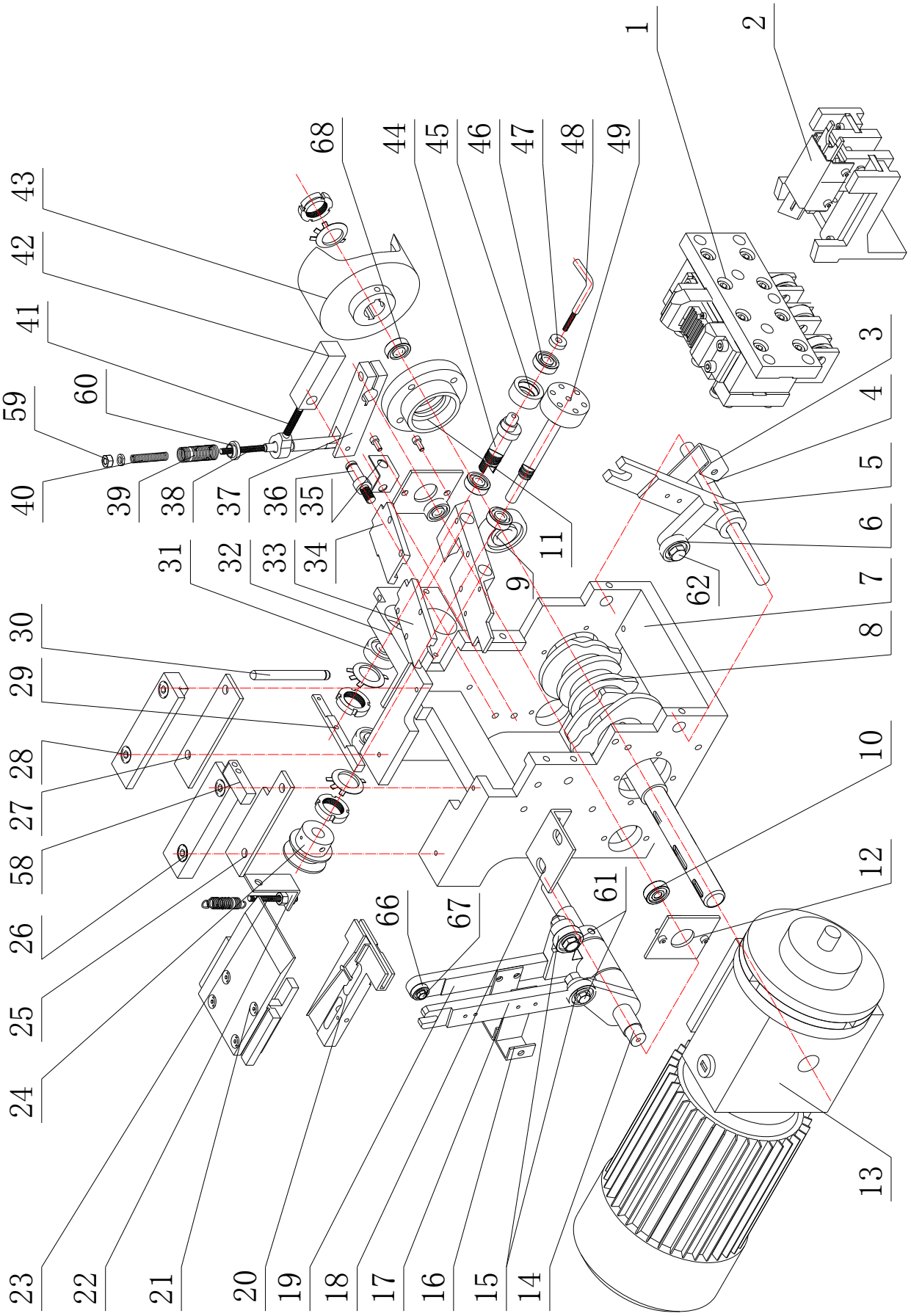
- Take down the tension springs of the left, middle and right bars.
- Move the upper slide table afterward slightly, and take down the left, middle and right bars.
- Wipe clean the left, middle and right bars and the inside of the cavities.
- Apply a little lighter lubricating oil around the left, middle and right bars.
- Reinstall the three bars and the springs.
- If the machine has been used for a long time, the cutter of the middle bar and the right partition plate fail to cut off the strap, you can move the middle bar towards the right partition plate, making them a little closer, but make sure that the middle bar can move up and down smoothly, or replace the middle bar or the right partition plate.

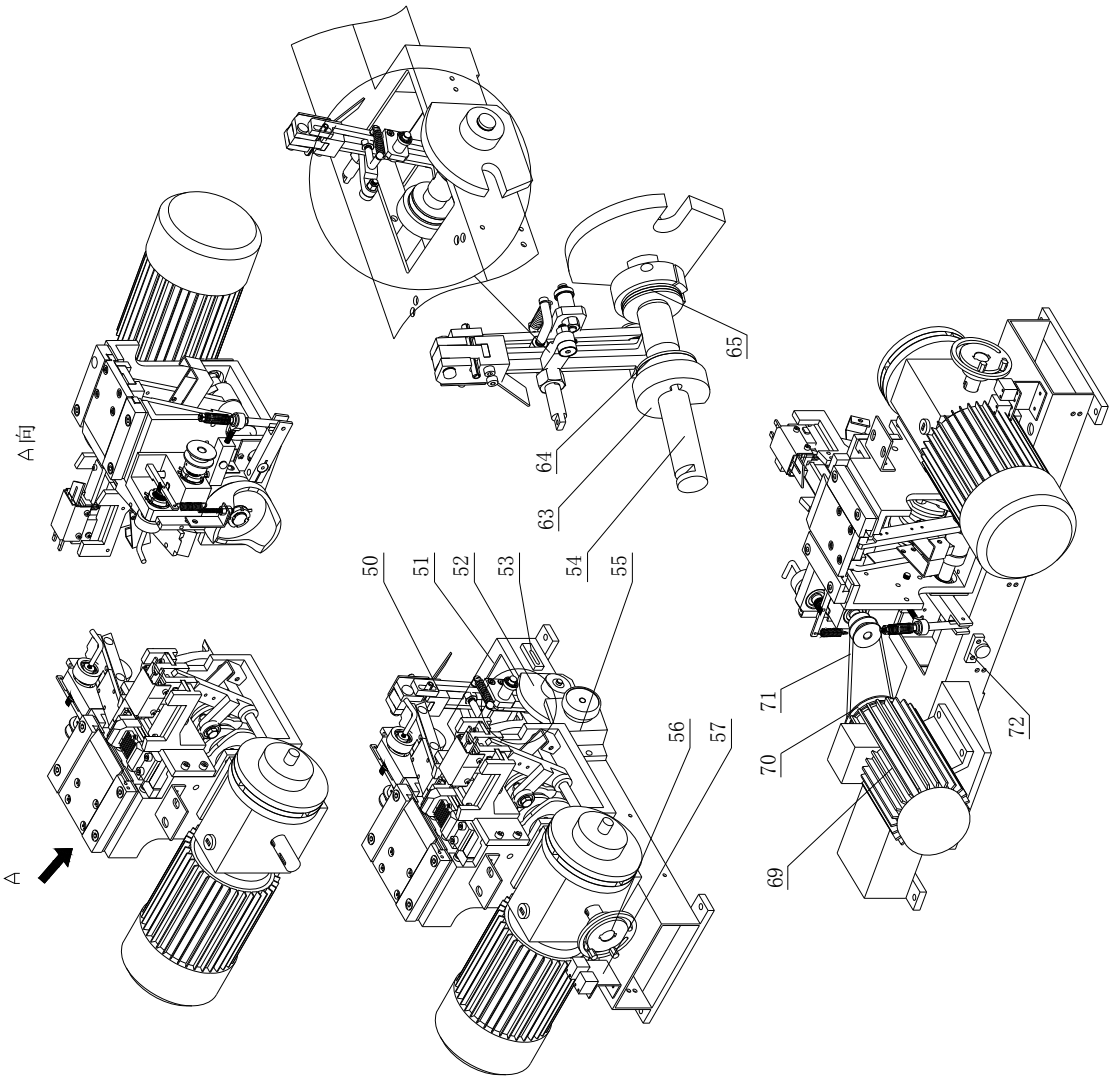
PART 7 MAIN PARTS DIAGRAMS AND LISTS



No.	Description	Q'TY	Remarks
1	Cabinet unit	1	
2	Panel 1	1	
3	Arch unit	1	
4	Panel 2	1	
5	Control panel	1	
6	Reel control unit	1	

7.1 Strapping head



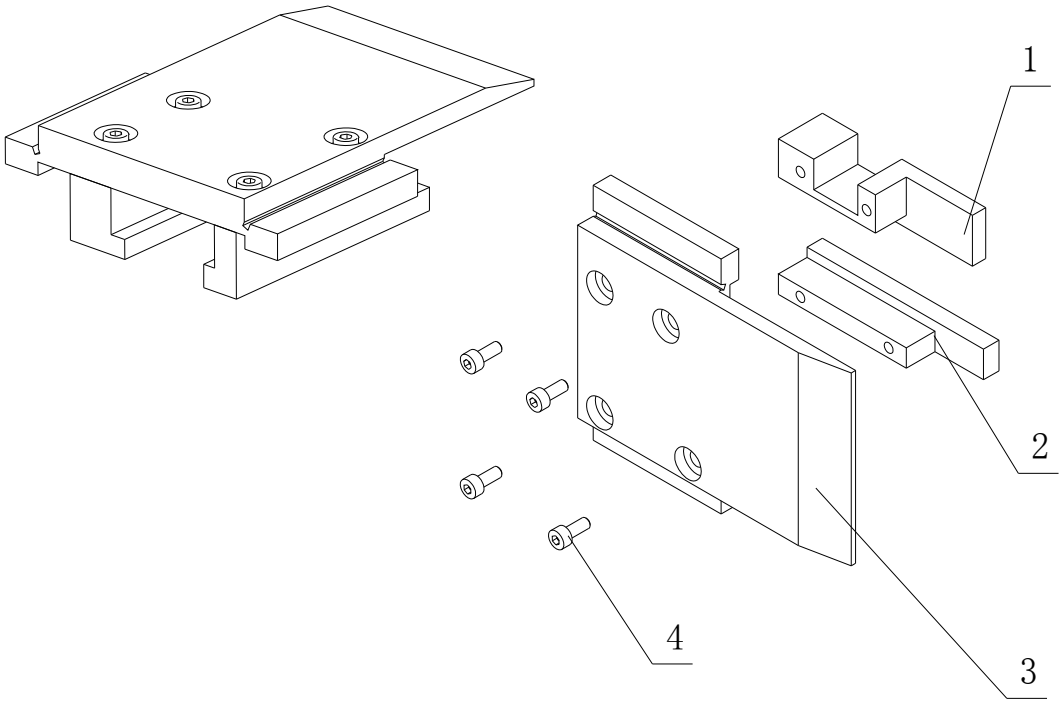

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Main Parts List

No.	Description	Code No.	Q'TY	Remarks
1	Bar guide assembly		1	See the diagram.
2	Heater assembly		1	See the diagram.
3	Hanging plate	01-144	1	
4	Shaft	01-131	1	
5	Heater crank	01-143	1	
6	Bearing	80100	1	Imported
7	Large bracket	01-1	1	
8	Parts of main shaft		1	See the diagram.
9	Bearing	80104	2	
10	Bearing	80102	2	
11	Bearing seat	01-14	2	
12	Baffle plate	01-18	2	
13	Motor and decelerator	YS8024B14 or YS7124B14	1	Decelerator NMRV40/30
14	Shaft	01-17	1	
15	Bearing	80100	2	
16	Hanging plate	01-144	2	
17	Guide plate crank	01-224	1	
18	Support plate	01-67	1	
		01-67A	1	For unmanned machine
19	Slide table crank	01-217	1	
20	Guide plate (T-shaped)	01-226	1	
21	Slide table assembly		1	See the diagram.
22	Bent plate	01-33	1	
23	Outer hexagonal screw	M5×50	1	Holed
24	Belt pulley	01-39	1	
25	Left guide plate	01-24	1	
26	Left guide rail	01-22	1	
27	Right guide plate	01-25	1	
28	Right guide rail	01-23	1	
29	Pry bar	01-32	1	
30	Ejector rod	01-51	1	
31	Bearing	50102	4	

32	Roller holder	01-1-2	1	
33	Front cover	01-56	1	
34	Rear cover	01-57	1	
35	Bent plate	01-40	1	
36	Shaft	01-50	1	
37	Clamping arm	01-200	1	
38	Pull bar	01-197	1	
39	Spring	01-136(1)	1	
40	Spring	01-136(2)	1	
41	Knuckle bearing	Φ8；SA8TK	1	
42	Pendulum block	01-196	1	
43	Edge cam	01-11	1	
44	Eccentric shaft	01-35	1	
45	Roller	01-36	1	
46	Bearing	80100	1	
47	Pressing ring	01-38	1	
48	Wrench	01-42	1	
49	Roller shaft	01-37	1	
50	Tension crank		1	See the diagram.
51	Channel steel	101A-37	1	
52	Positioning plate	01-68	1	
53	Limit plate	01-69	1	
54	Shaft	01-45	1	
55	Tension adjusting device		1	See the diagram.
56	Probe holder	01-60	1	
57	Switch bracket	01-65	1	
58	Block	01-44	1	
59	Nut	Φ6	1	
60	Spring base	01-52	1	
61	Shaft	01-218	2	
62	Shaft	01-146	1	
63	Stop collar	01-47	2	
64	Bearing	8104	1	
65	Bearing	889104	1	
66	Bearing	80017	1	
67	Shaft	01-219	1	
68	Bearing	8204	1	
69	Motor	YS6334B3	2	
70	Belt pulley	00-25	1	
71	Belt	M21	1	
72	Limit plate	01-69	1	

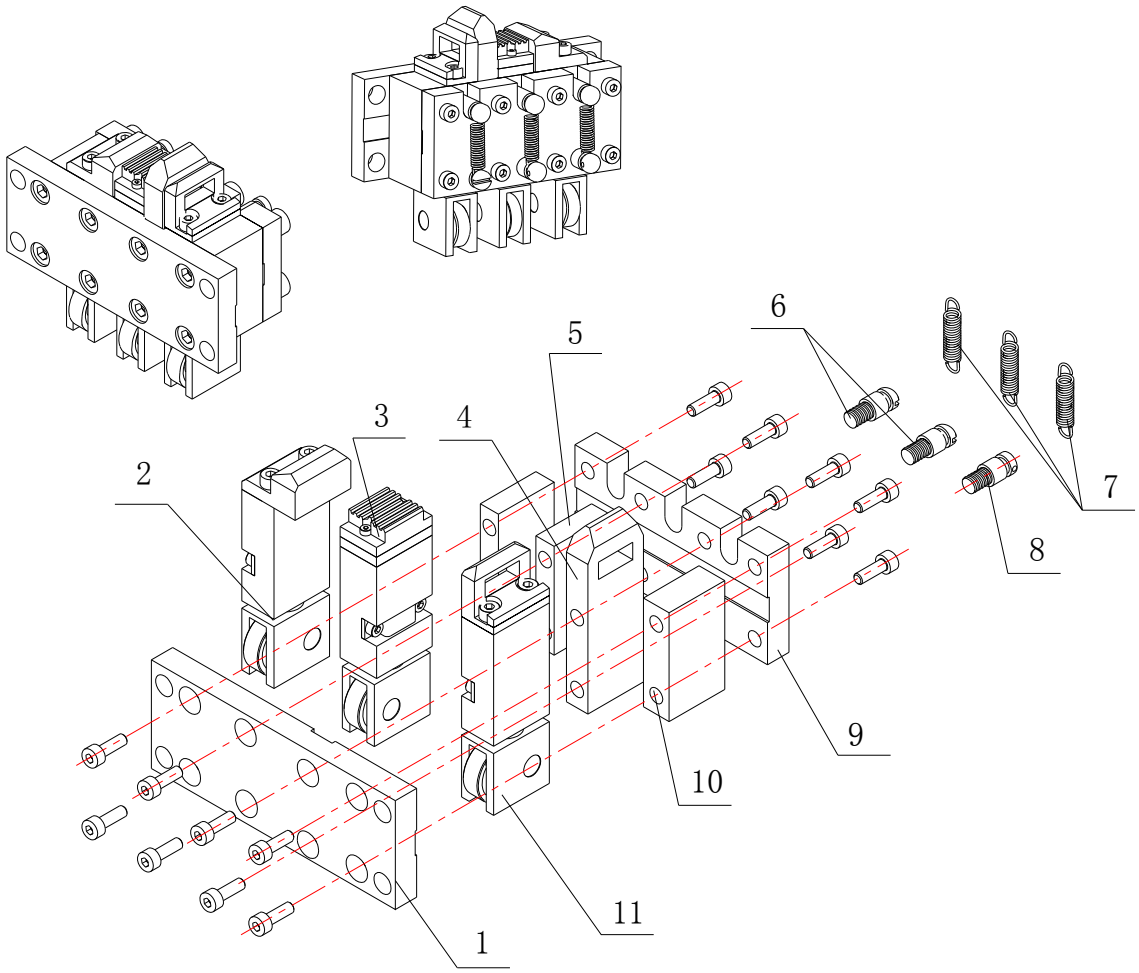
7.1.1 Slide table assembly



Main Parts List

No.	Description	Code No.	Q'TY	Remarks
1	Right rail of guide plate	01-27	1	
2	Left rail of guide plate	01-26	1	
3	Slide table	01-20	1	
4	Screw	M5×16	4	

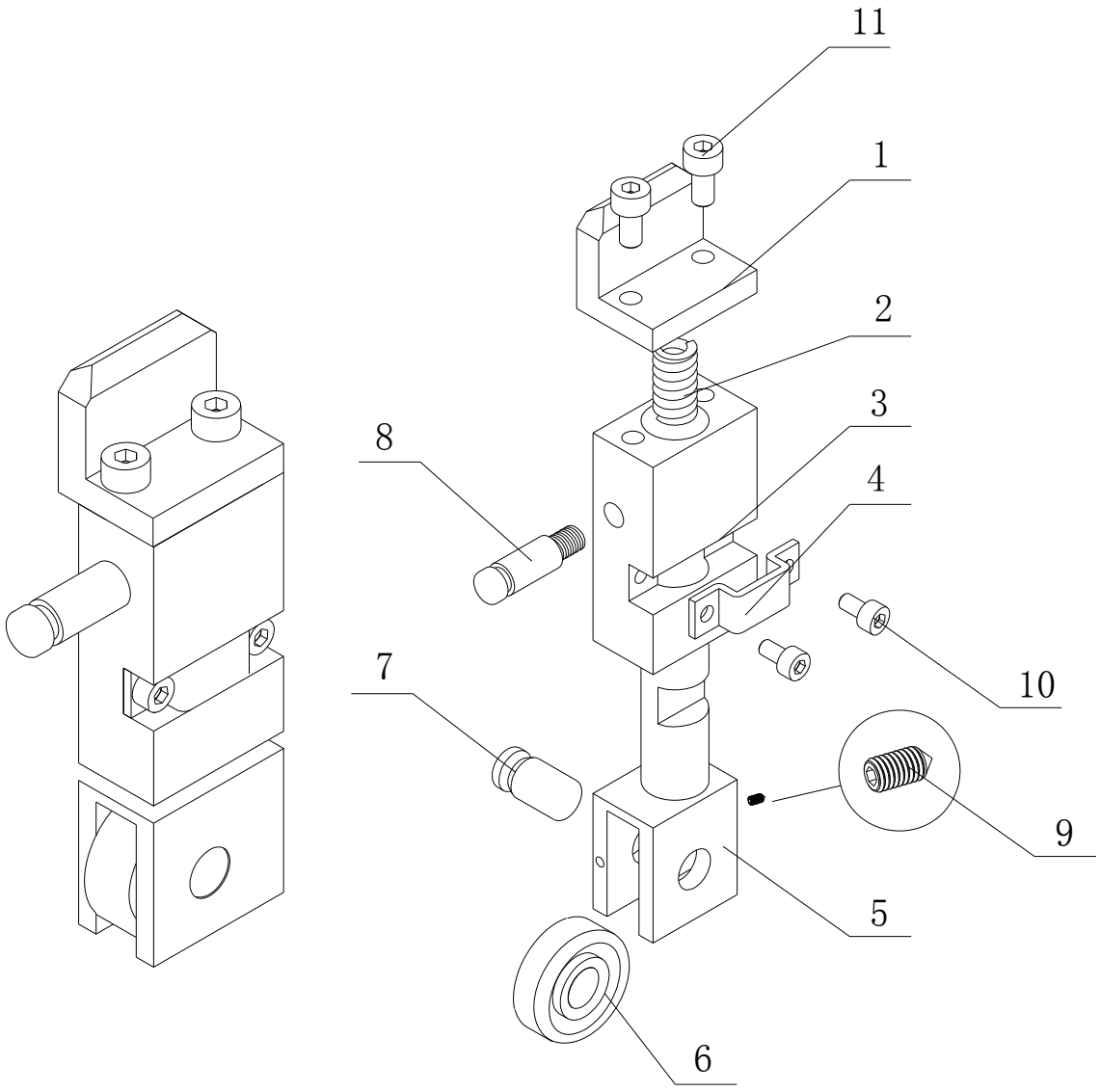
7.1.2 Bar guide assembly



Main Parts List

No.	Description	Code No.	Q'TY	Remarks
1	Front plate	01-90	1	
2	Left bar assembly		1	See the diagram.
3	Middle bar assembly		1	See the diagram.
4	Right partition board	01-93	1	
5	Left partition board	01-92	1	
6	Post	01-96	2	
7	Tension spring	01-97	3	
8	Post	01-95	1	
9	Back plate	01-91	1	
10	Baffle plate	01-94	2	
11	Right bar assembly		1	See the diagram.

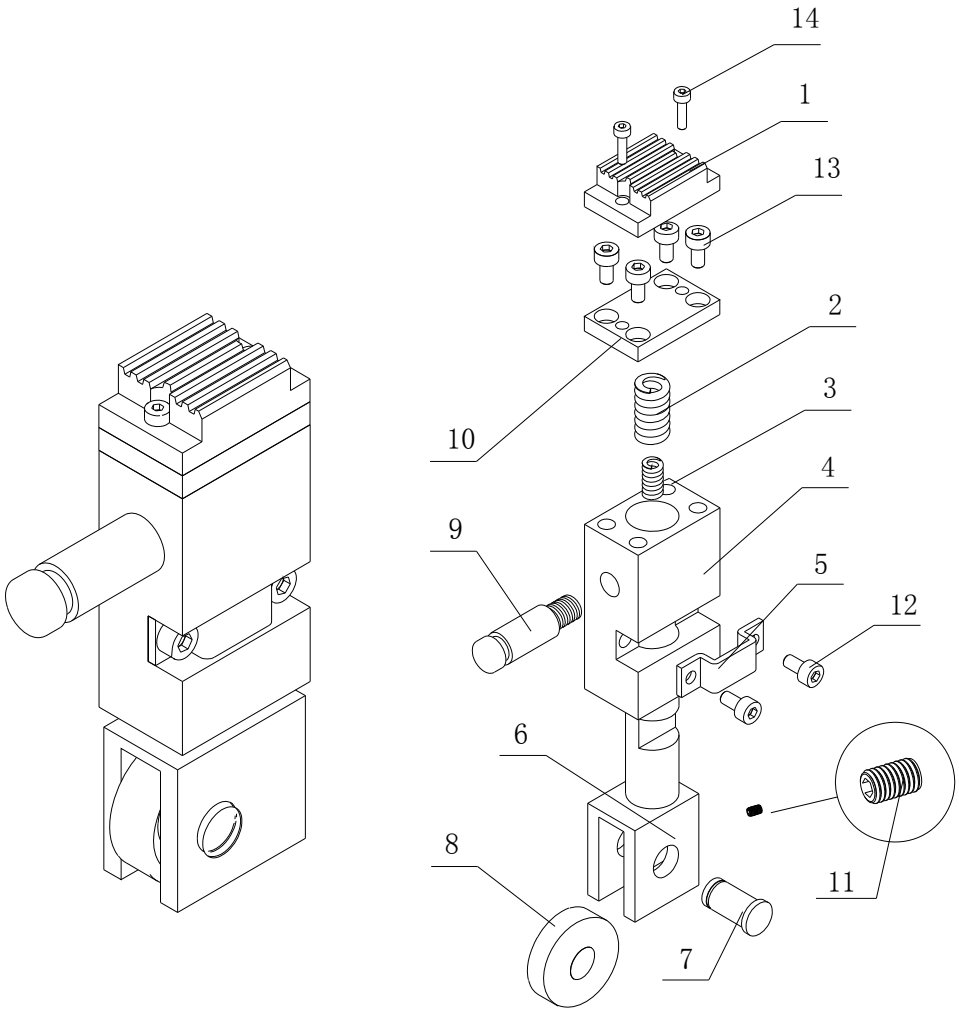
7.1.2.1 Left bar assembly



Main Parts List

No.	Description	Code No.	Q'TY	Remarks
1	Left jack catch	01-112	1	
2	Compression spring	01-103	1	
3	Cavity	01-101	1	
4	Stop dog	01-105	1	
5	Spring base	01-100	1	
6	Bearing	80100	1	
7	Small shaft	01-106	1	
8	Post	01-109	1	
9	Set screw	M5×6	1	Cusp
10	Inner hexagon screw	M4×12	2	
11	Inner hexagon screw	M5×20	2	

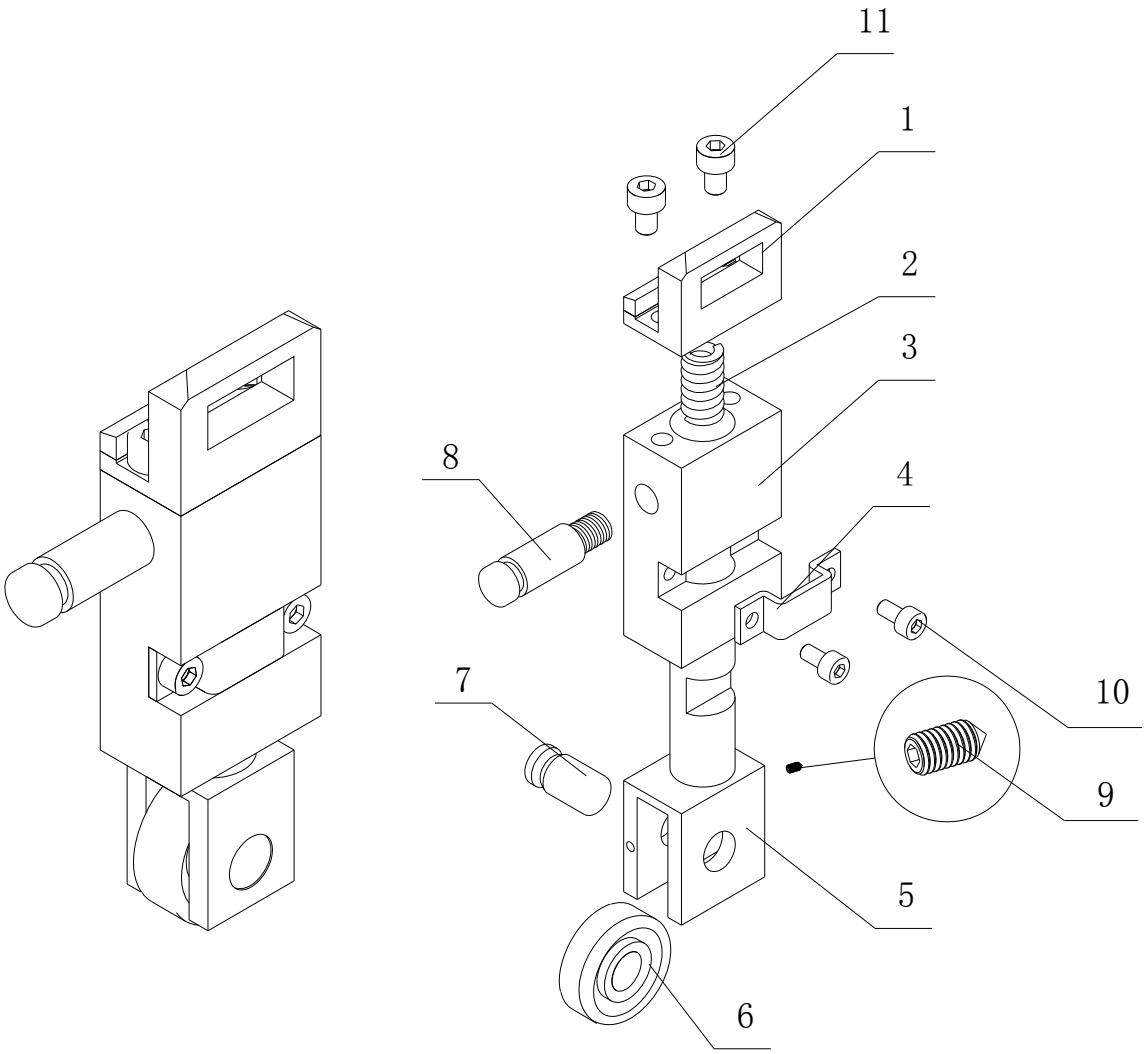
7.1.2.2 Middle bar assembly



Main Parts List

No.	Description	Code No.	Q'TY	Remarks
1	Bar	01-114	1	
2	Compression spring	01-103	1	
3	Compression spring	01-104	1	
4	Cavity	01-102	1	
5	Stop dog	01-105	1	
6	Spring base	01-100	1	
7	Shaft	01-107	1	
8	Roller	01-108	1	
9	Post	01-109	1	
10	Base plate	01-113	1	
11	Set screw	M5×6	1	Flattop
12	Inner hexagon screw	M4×12	2	
13	Inner hexagon screw	M4×12	4	
14	Inner hexagon screw	M5×12	2	

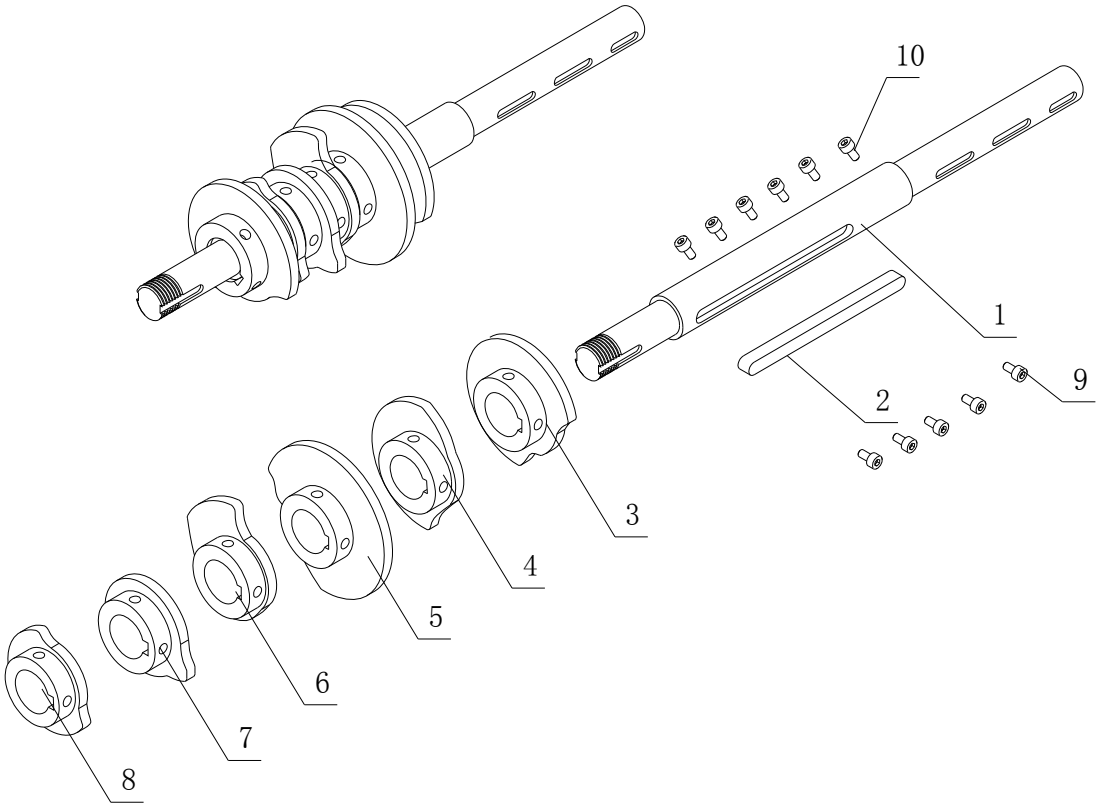
7.1.2.3 Right bar assembly



Main Parts List

No.	Description	Code No.	Q'TY	Remarks
1	Right jack catch	01-115	1	
2	Compression spring	01-104	1	
3	Cavity	01-101	1	
4	Stop dog	01-105	1	
5	Spring base	01-100	1	
6	Bearing	80100	1	
7	Small shaft	01-106	1	
8	Post	01-109	1	
9	Set screw	M5×6	1	Cusp
10	Inner hexagon screw	M4×12	2	
11	Inner hexagon screw	M5×16	2	

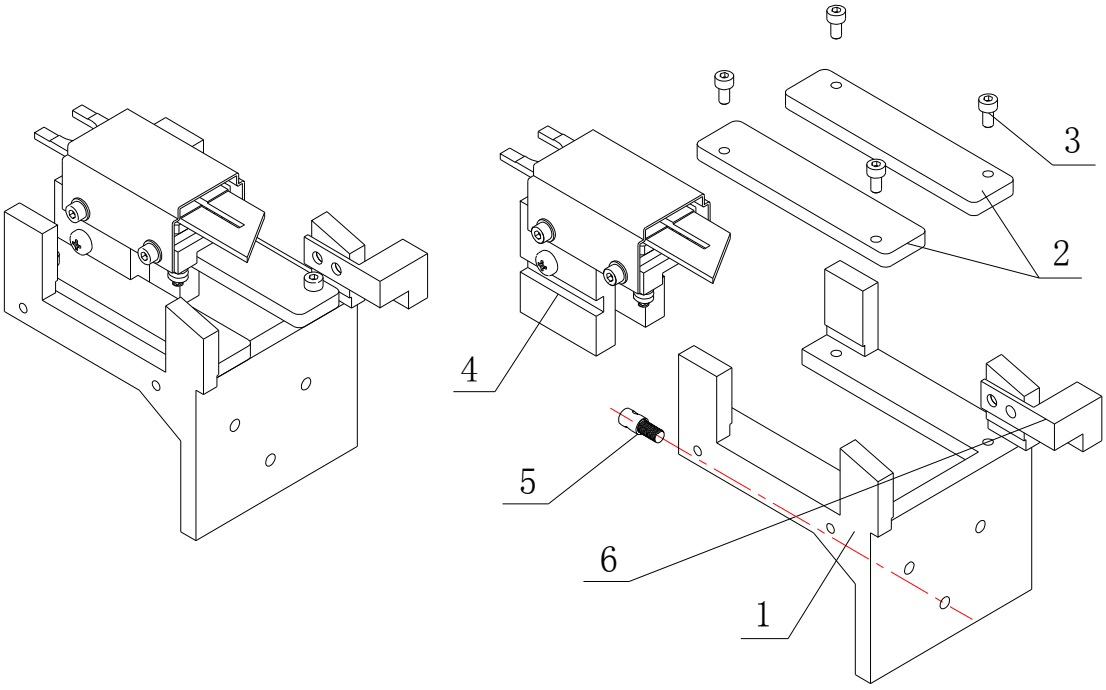
7.1.3 Parts of main shaft



Main Parts List

No.	Description	Code No.	Q'TY	Remarks
1	Main shaft	01-12 (1)	1	
2	Flat key		1	8×115
3	Cam	01-5	1	
4	Cam	01-6	1	
5	Cam	01-7	1	
6	Cam	01-8	1	
7	Cam	01-9	1	
8	Cam	01-10	1	
9	Screw	M8×8	6	
10	Screw	M8×10	6	

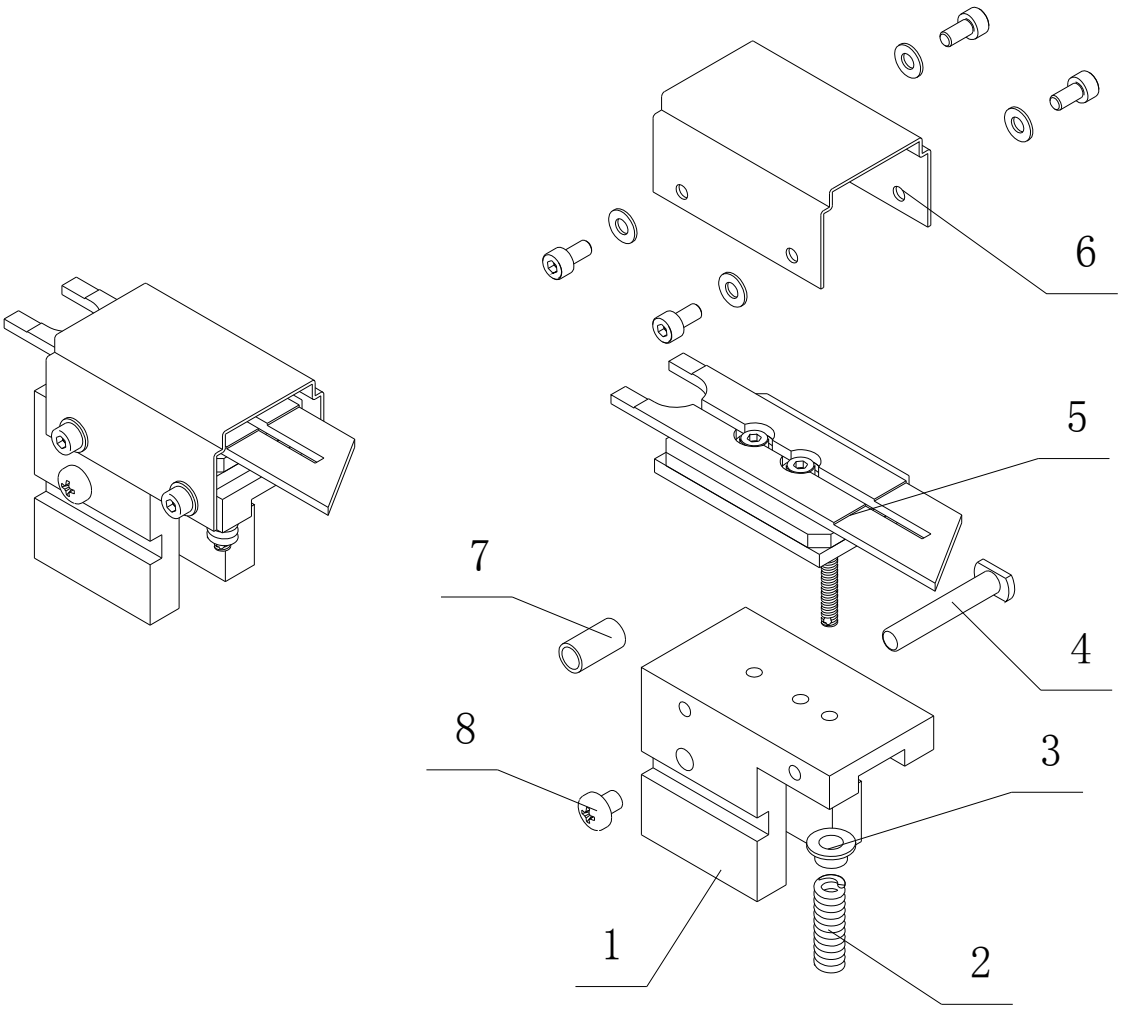
7.1.4 Heater assembly



Main Parts List

No.	Description	Code No.	Q'TY	Remarks
1	Bracket	01-130	1	
2	Guide plate	01-132	2	
3	Inner hexagon screw	M5×16	4	
4	Assembly of heater plate and slide table		1	See the diagram.
5	Small shaft	01-133	1	
6	Block	01-43	1	

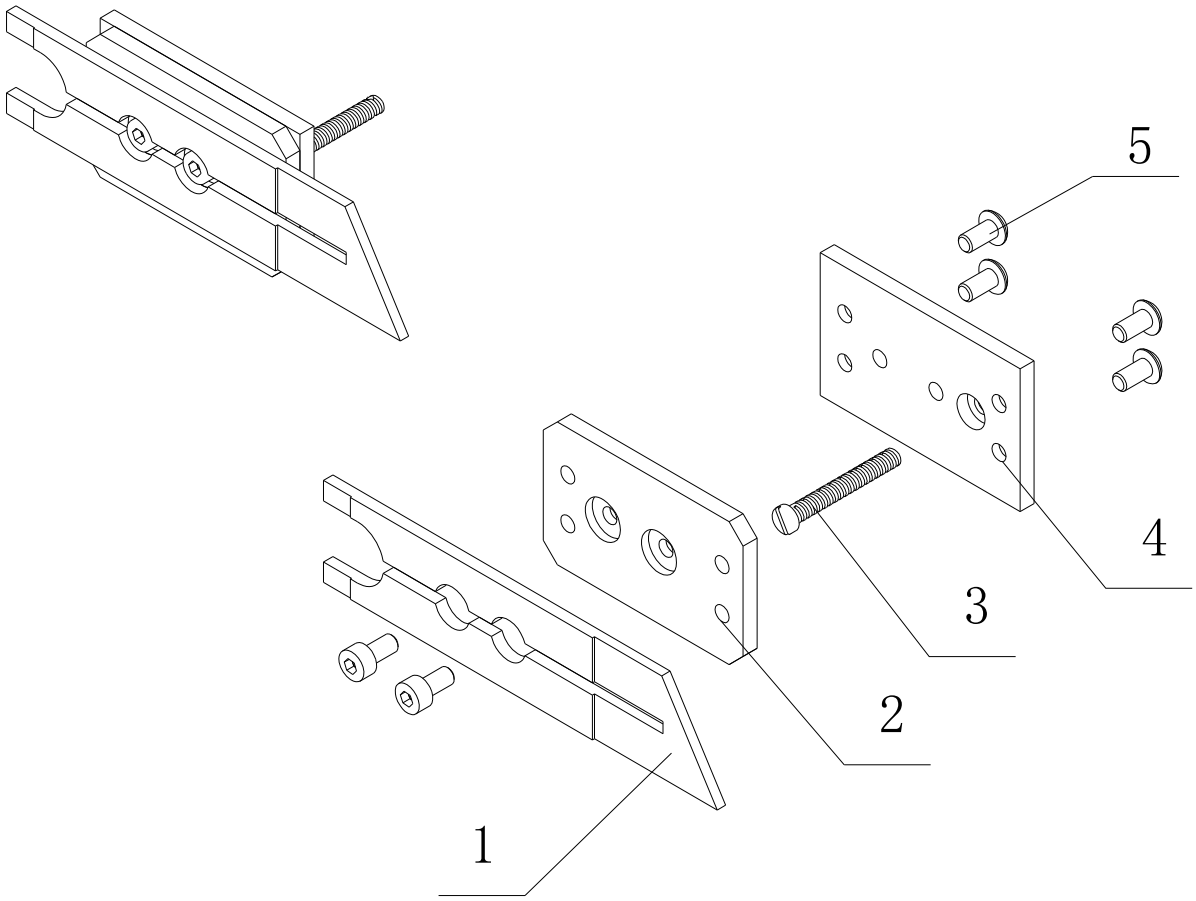
7.1.4.1 Assembly of heater plate and slide table



Main Parts List

No.	Description	Code No.	Q'TY	Remarks
1	Slider	01-137	1	
2	Compression spring	01-140	1	
3	Spring base	01-141	1	
4	Shaft	01-138	1	
5	Heater plate assembly		1	See the diagram.
6	Cover	01-142	1	
7	Shaft sleeve	01-139	1	
8	Phillips screws	M4×10	1	

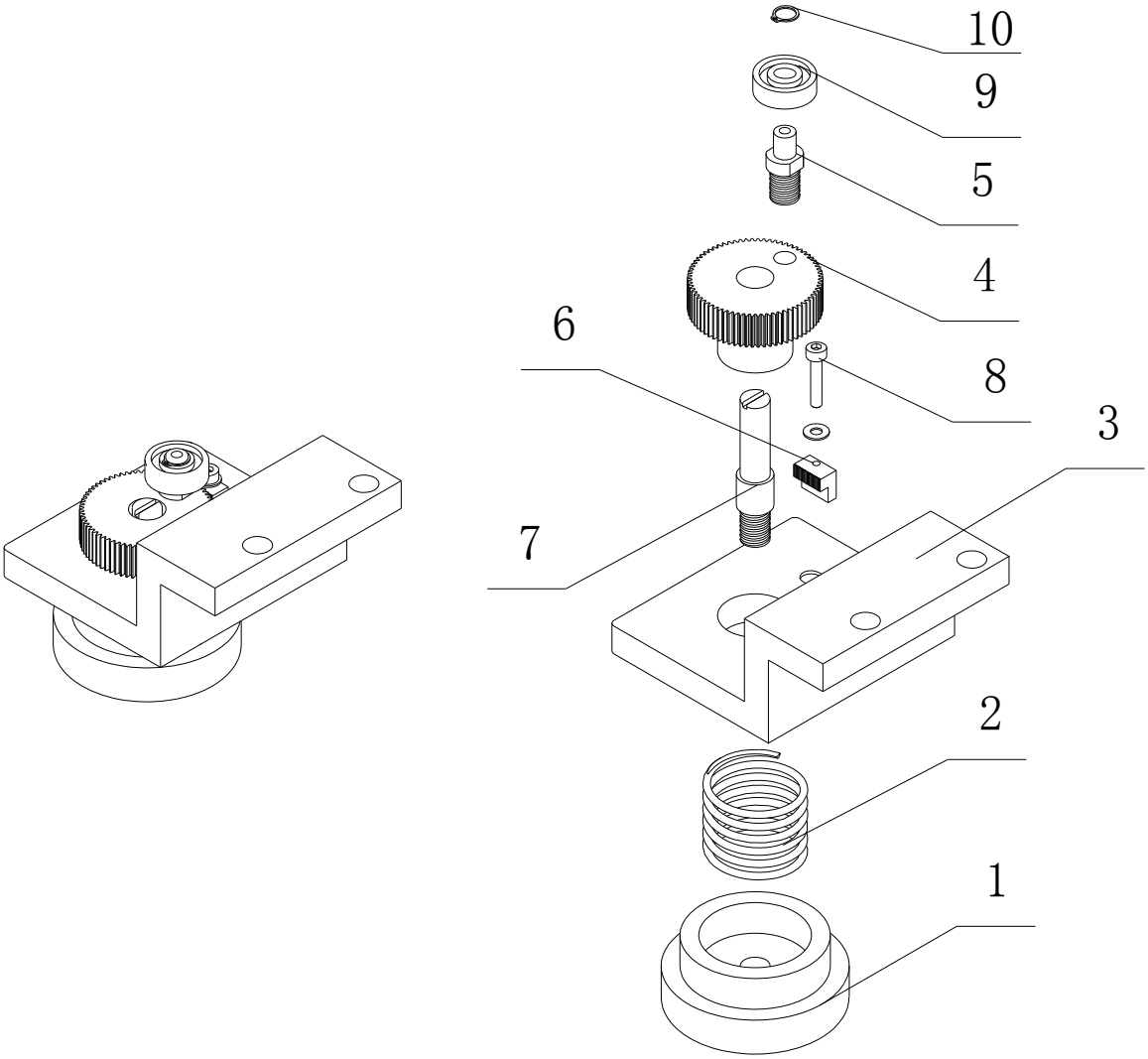
⑤ Heater plate assembly



Main Parts List

No.	Description	Code No.	Q'TY	Remarks
1	Heater plate	01-153	1	
2	Insulation board	01-150	1	
3	Bolt	01-152	1	
4	Base plate	01-151	1	
5	Phillips screws	M4×10	4	

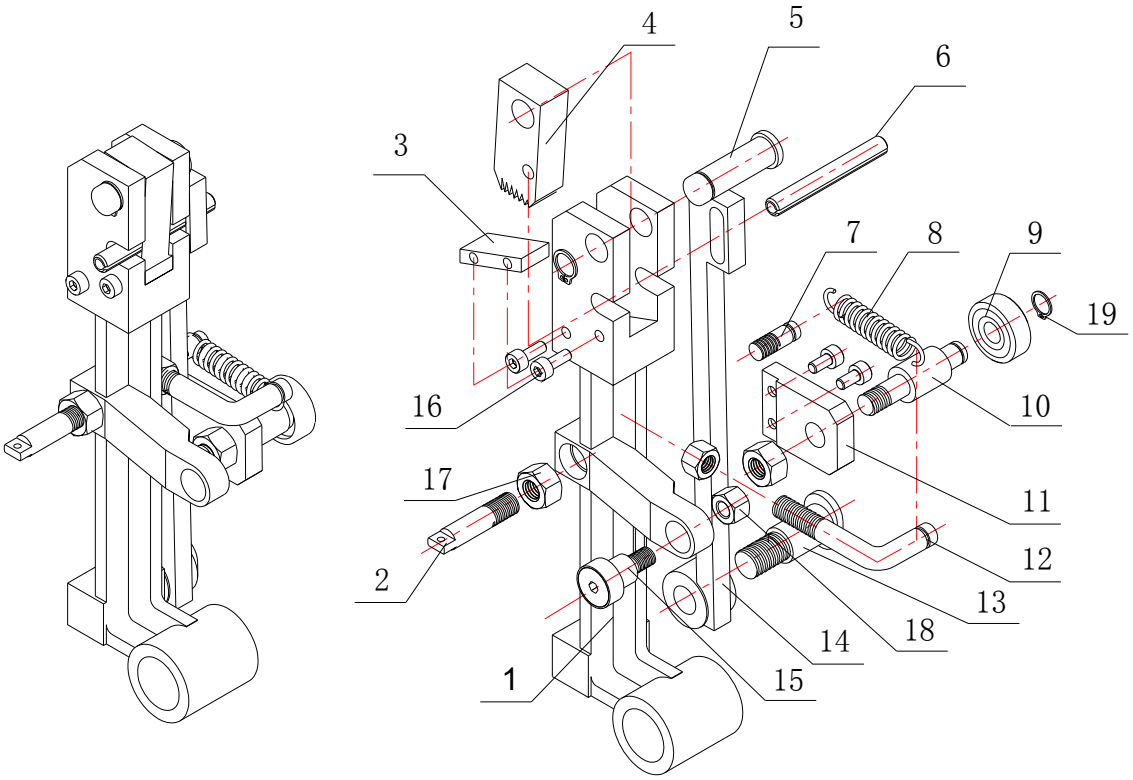
7.1.5 Tension adjusting device



Main Parts List

No.	Description	Code No.	Q'TY	Remarks
1	Knob	01-205	1	
2	Compression spring	01-207	1	
3	Bracket	01-204	1	
4	Gear	01-209	1	
5	Small shaft	01-210	1	
6	Spur rack	01-211	1	
7	Shaft	01-206	1	
8	Inner hexagon screw	M4×16	1	
9	Bearing	80026	1	
10	Stop collar	Φ4	1	

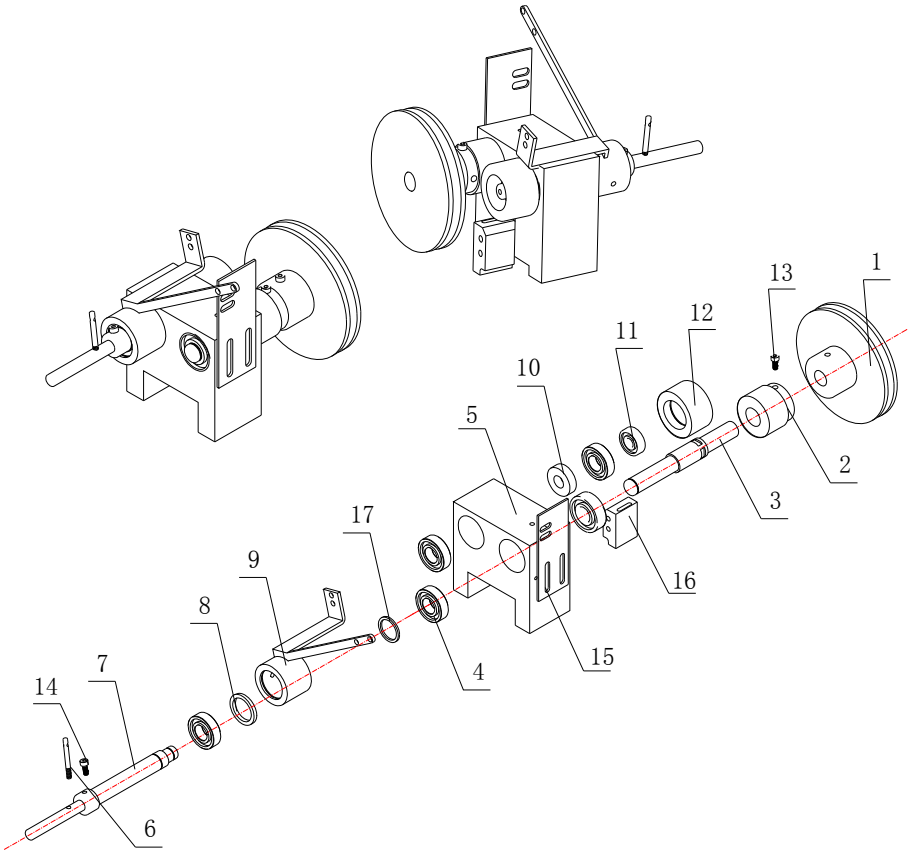
7.1.6 Tension crank



Main Parts List

No.	Description	Code No.	Q'TY	Remarks
1	Crank	01-160	1	
2	Pull bar	01-169	1	
3	Base plate	01-160-1	1	
4	Strap clamp block	01-165	1	
5	Strap clamp shaft	01-166	1	
6	Pin	01-163	1	
7	Post	01-174	1	
8	Tension spring	01-162	1	
9	Bearing	80026	1	
10	Small shaft	01-176	1	
11	Pedestal	01-175	1	
12	Bent rod	01-161	1	
13	Rod shaft	01-168	1	
14	Rod	01-177	1	
15	Bearing	CFØ10	1	
16	Inner hexagon screw	M4×14	2	
17	Nut	Φ10	1	
18	Nut	Φ6	2	
19	Stop collar	Φ4	2	

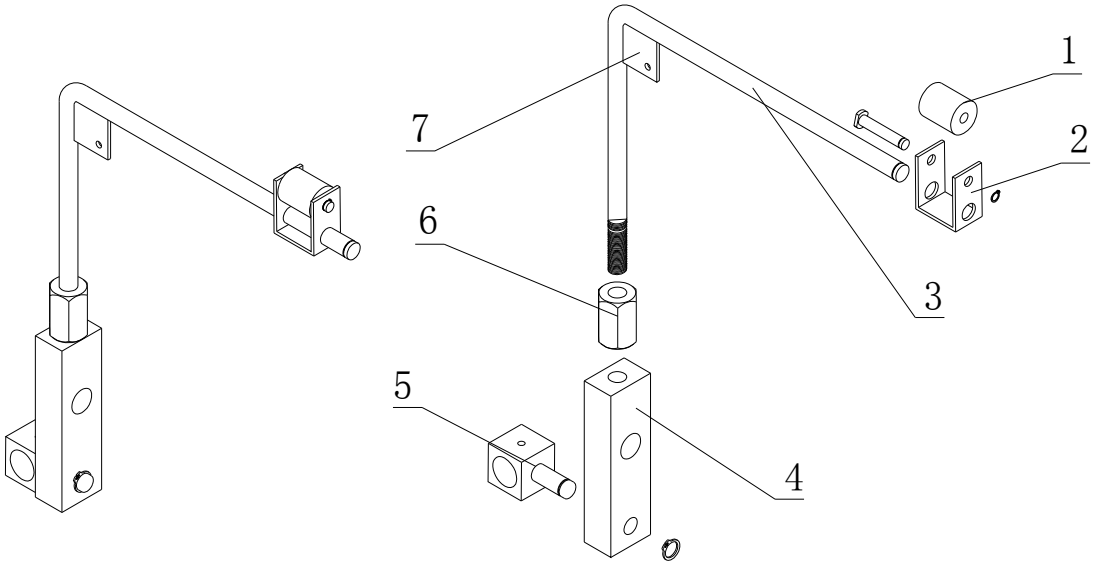
7.2 Strap pre-feeding mechanism



Main Parts List

No.	Description	Code No.	Q'TY	Remarks
1	Belt pulley	00-92	1	
2	Upper strap conveying roller	00-95	1	
3	Upper shaft	00-93	1	
4	Bearing	80102	6	
5	Base	00-83	1	
6	Oscillating bar	00-97	1	
7	Lower shaft	00-94	1	
8	Spacer bush	00-88	1	
9	Rocker roller wheel	00-85; 00-86	1	Welded together
10	Spacer bush	00-84	1	
11	Bearing	80101	1	
12	Lower strap conveying roller	01-98	1	
13	Set screw	M6×10	1	
14	Set screw	M5×6	1	
15	Switch bracket	00-89	1	
16	Strap inlet	00-108	1	
17	Spacer bush	00-87	1	

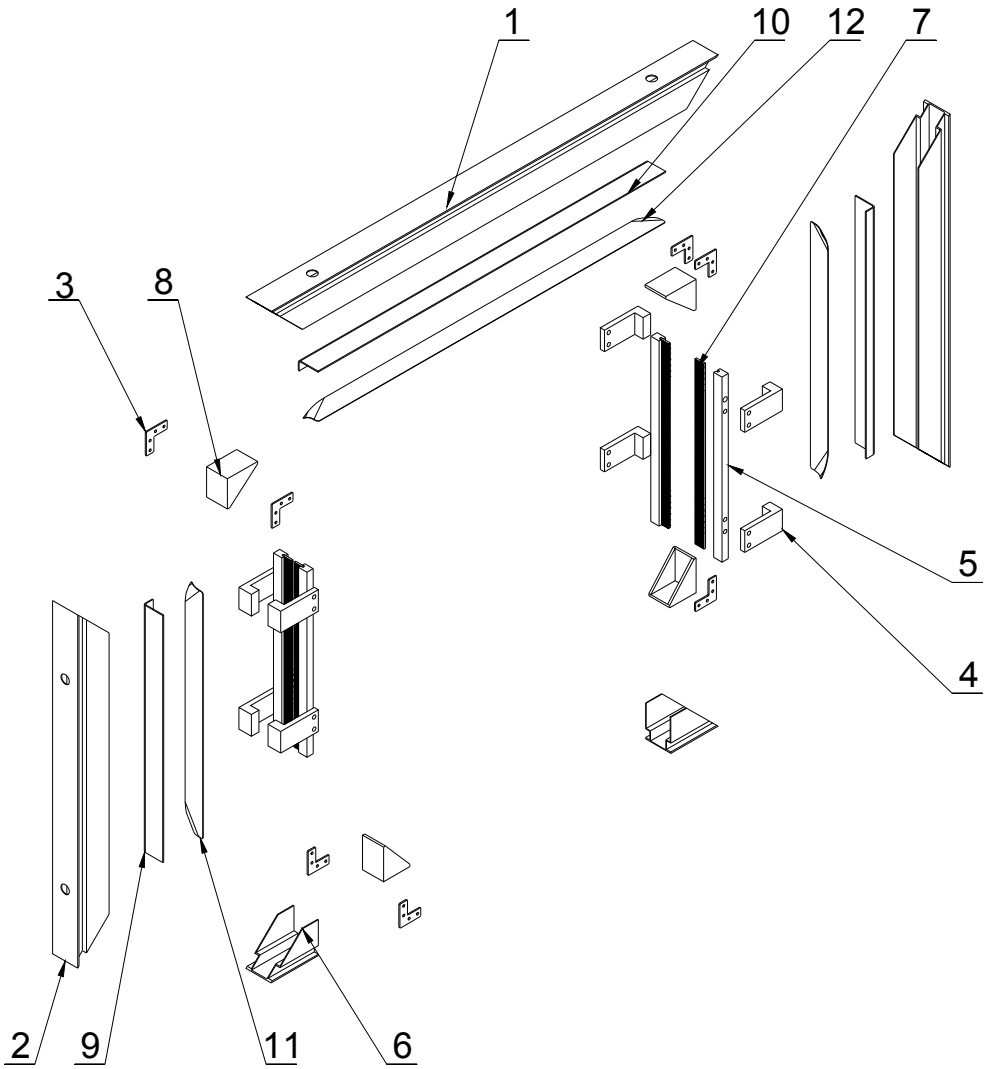
7.3 Swing mechanism



Main Parts List

No.	Description	Code No.	Q'TY	Remarks
1	Roller	00-134	1	
2	Roller holder	00-133	1	
3	Oscillating bar	00-130(1)	1	
4	Oscillating block	00-127	1	
5	Belt clamp	00-128	1	
6	Nut	00-129	1	
7	Spring hook piece	00-132	1	

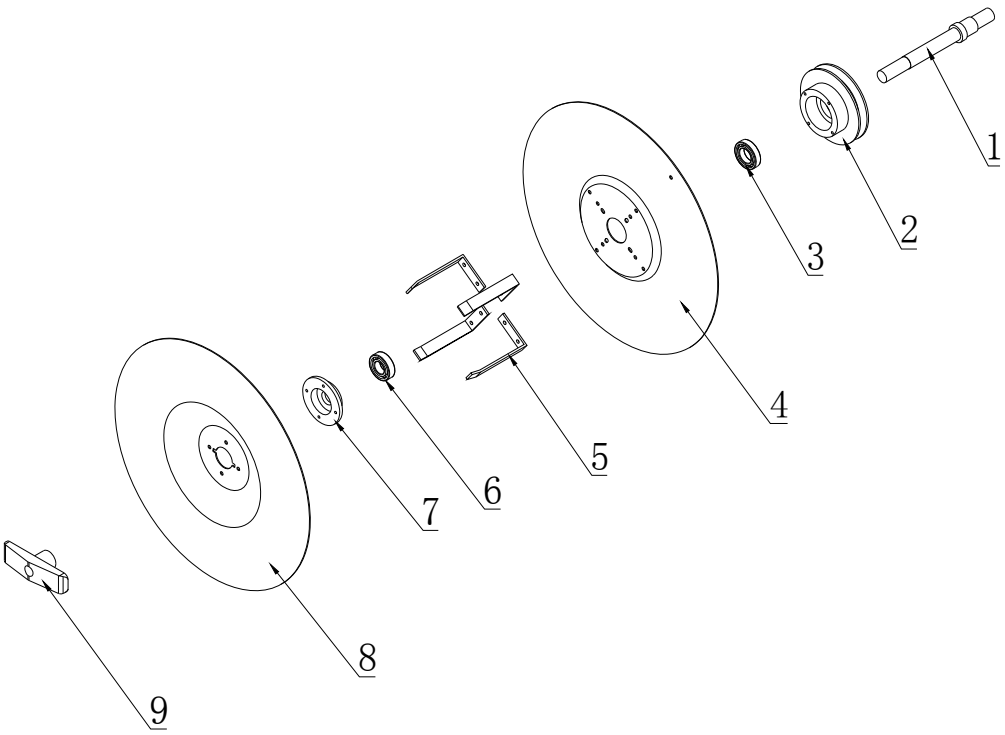
7.4 Arch unit



Main Parts List

No.	Description	Code No.	Q'TY	Remarks
1	Beam	MH-A-1	1	
2	Upright post	MH-A-2	2	
3	Connecting block	KG-02	8	
4	Plastic brush holder		8	
5	Brush holder	MH-D-1	4	
6	Foot piece	MH-A-3	2	
7	Brush	MH-D-2	4	
8	Plastic turn angle		4	
9	Upright base plate	MH-B-1	2	
10	Horizontal base plate	MH-B-2	1	
11	Turning plate	KF-00	2	
12	Turning plate	KF-00	3	

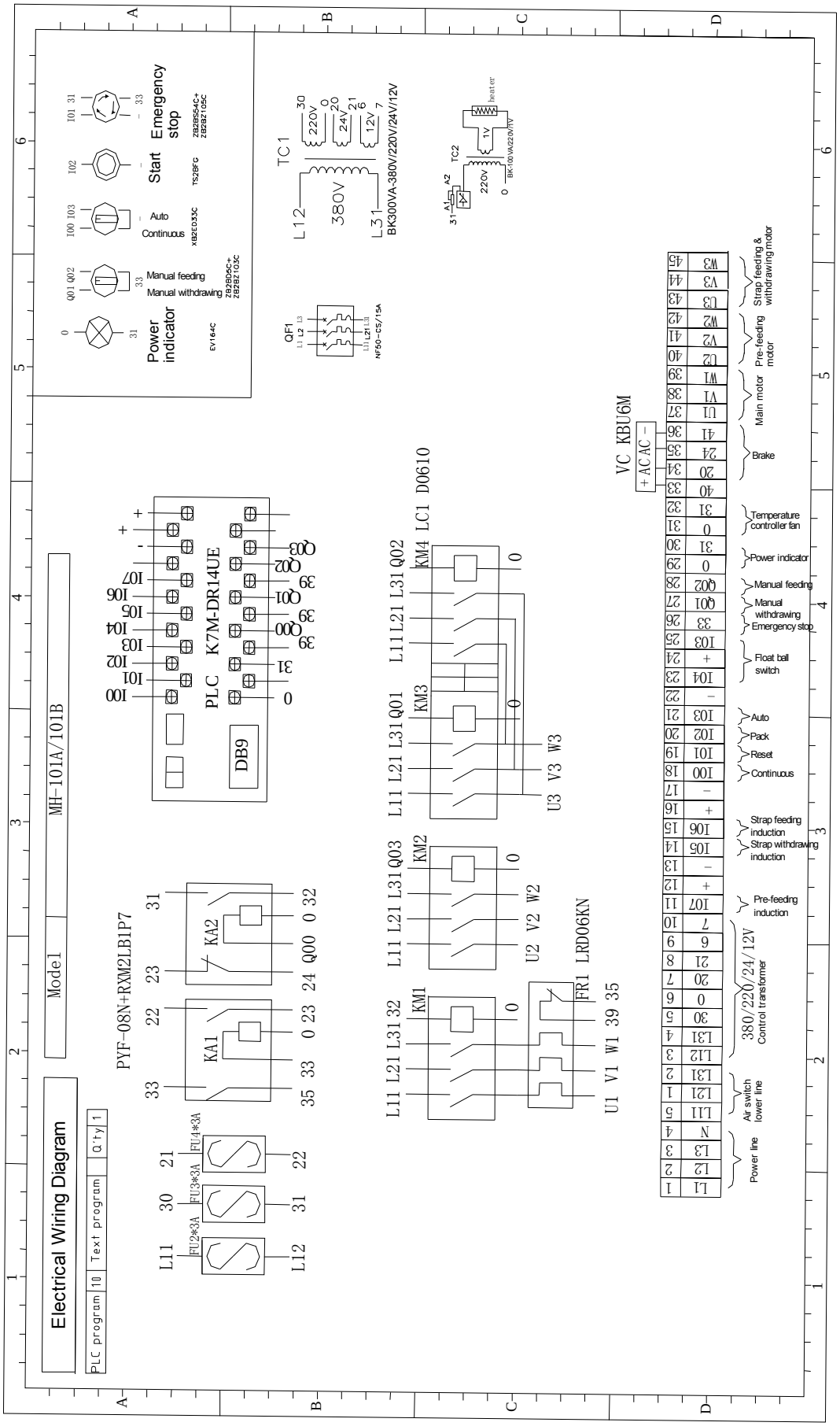
7.5 Strap control unit



Main Parts List

No.	Description	Code No.	Q'TY	Remarks
1	Shaft	00-114	1	
2	Brake wheel	00-111	1	
3	Deep groove ball bearing	6006	1	
4	Inner flange	00-142	1	
5	Support claw	00-143	1	
6	Deep groove ball bearing	6205	1	
7	Bearing seat	00-188	1	
8	Outer flange	00-189	1	
9	Handle	00-192	1	Iron/plastic

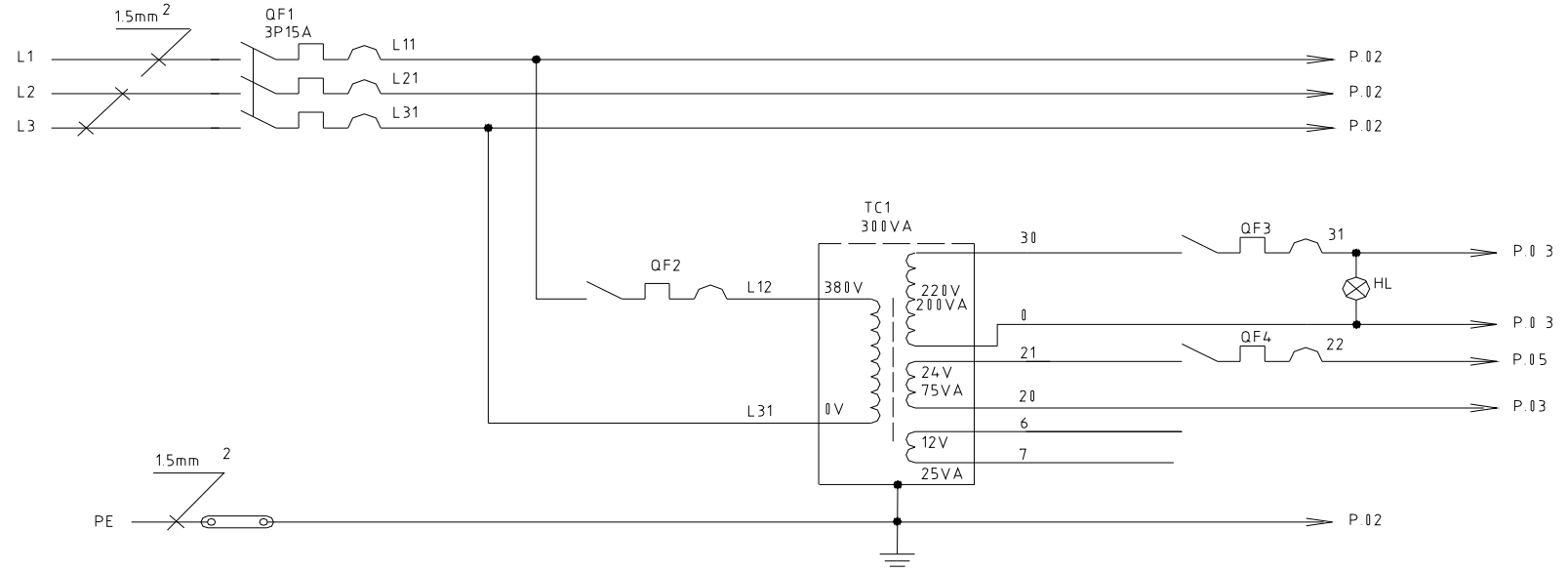
PART 8 MAIN ELECTRICAL WIRING AND SCHEMATIC DIAGRAMS



Main Power Supply and Control Power Supply

Model		MH-101A/101B
Breaker	QF1	15A/3P
Breaker	QF2	3A
	QF3,4	3A
	QF5	3A
Transformer	TC	300VA IN AC380V OUT AC220/24V
Total power		1.17KW

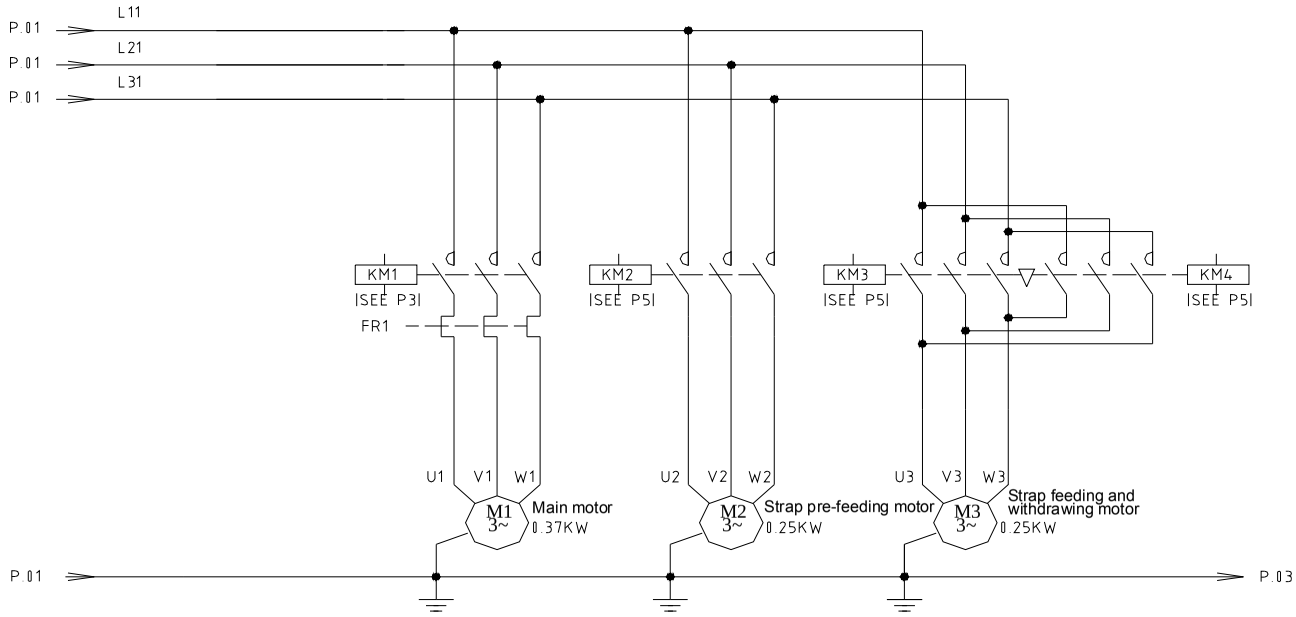
Main Input Power Supply
3Φ380V50HZ



Rev.	Approved Date	Remarks	DRAWN	CHECKED		
			DESIGNED	APPROVED	SCALE	TITLE: Main Power Supply and Control Power Supply
					P 01	DRAWING NO: MH-101A/101B Strapping Machine
						ISSUED DATE:

Motor Start

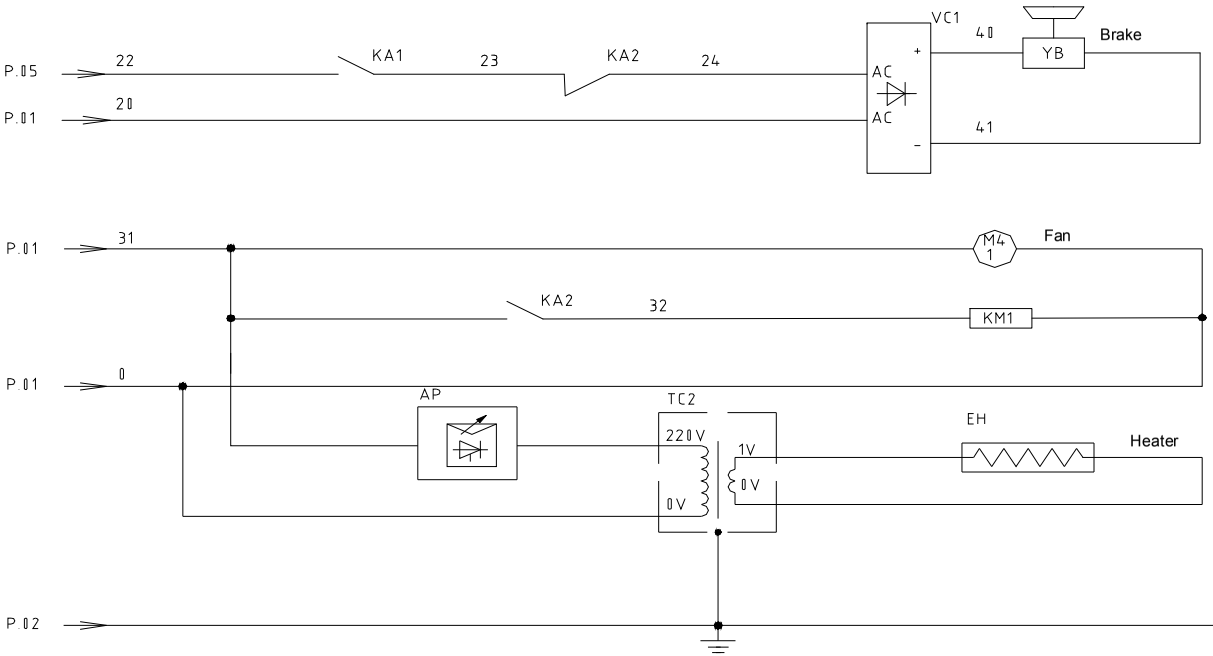
Model	MH-101A/101B	
Main motor	M1	380VAC/3/50HZ/0.37KW
Strap pre-feeding motor	M2	380VAC/3/50HZ/0.25KW
Strap feeding and withdrawing motor	M3	380VAC/3/50HZ/0.25KW
Overload relay	FR1,FR2	1-1.6A



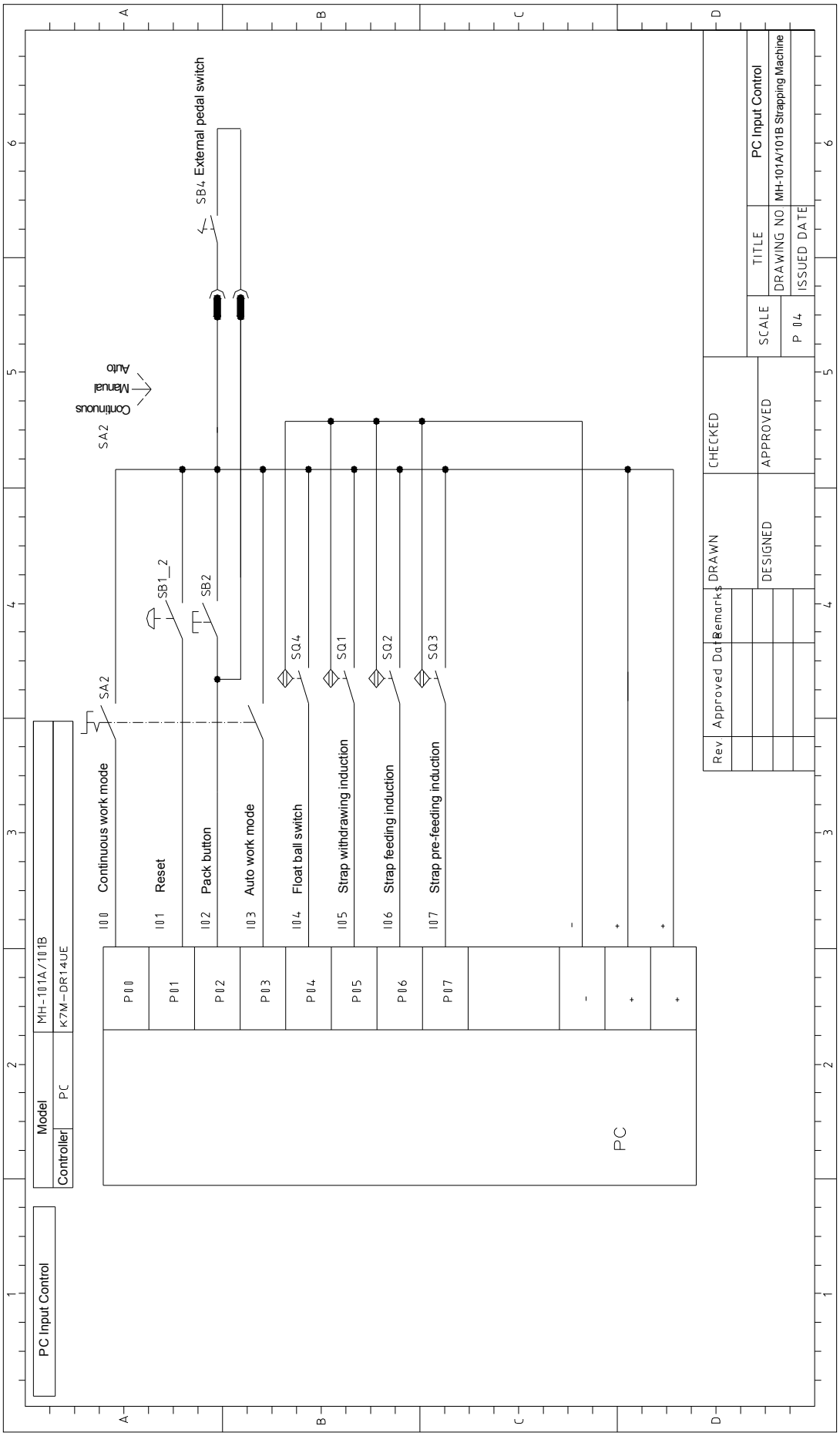
Rev.	Approved Date	Remarks	DRAWN	CHECKED		
			DESIGNED	APPROVED	SCALE	TITLE: Motor Start
					P.02	DRAWING NO.: MH-101A/101B Strapping Machine
						ISSUED DATE:

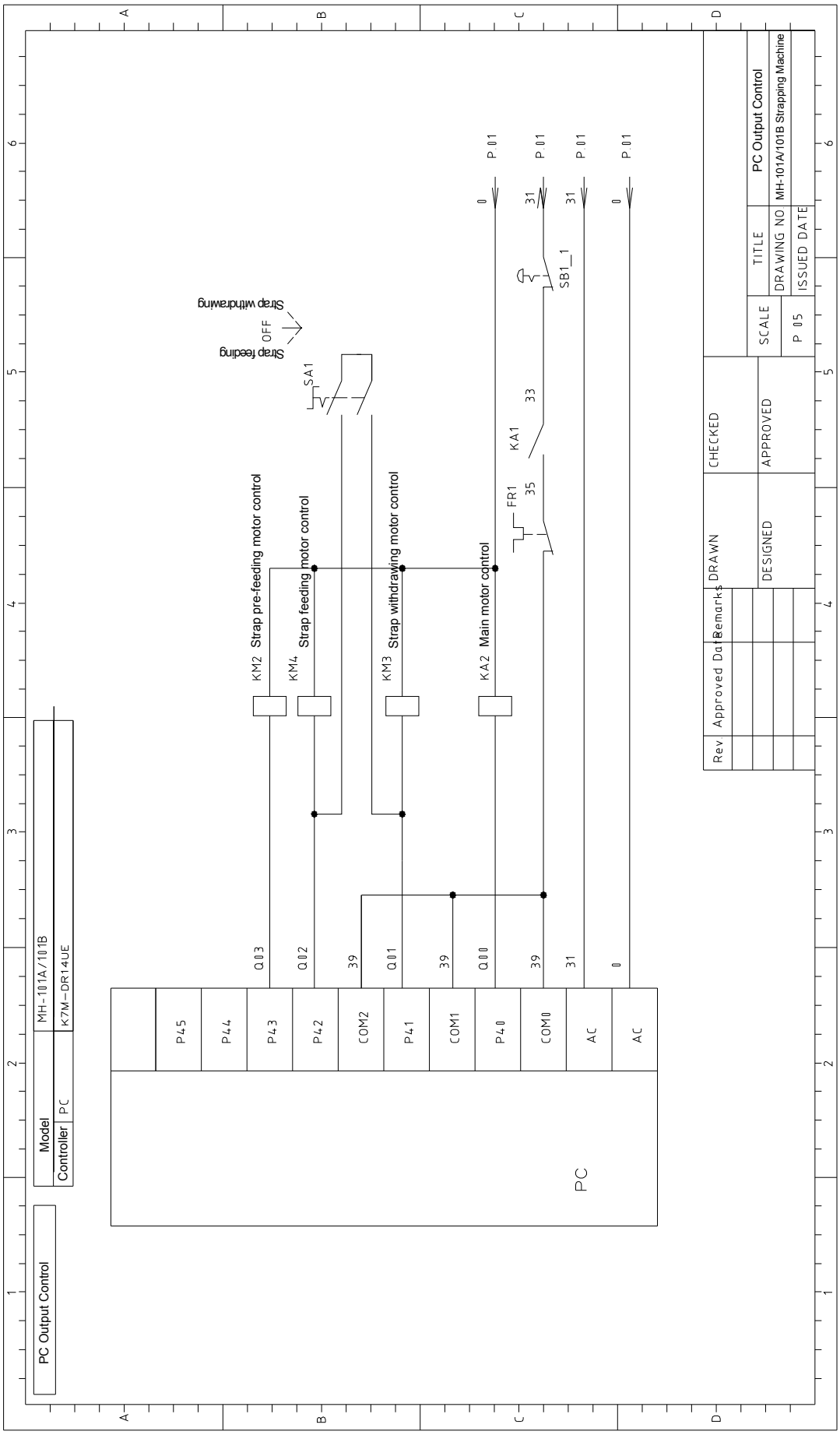
Brake and Heating

Model		MH-101A/101B
Brake	YB	DC24V 0.7A
Rectifier	VC	KBU6M
Big fan	M4	220VAC
Transformer	TC2	100VA IN.AC220V OUT.AC1V
Temperature controller	AP	0-200VAC, 0-300℃



Rev.	Approved Date	Remarks	DRAWN	CHECKED		
			DESIGNED	APPROVED	SCALE	TITLE: Brake and Heating
					P 03	DRAWING NO: MH-101A/101B Strapping Machine
						ISSUED DATE:


PC Input Control
Model
Controller
PC
MH-101A/101B
K7M-DR14UE
SB4 External pedal switch
Continuous
Manual
Auto
SA2
SB1_2
SB2
SQ4
SQ1
SQ2
SQ3
I00 Continuous work mode
I01 Reset
I02 Pack button
I03 Auto work mode
I04 Float ball switch
I05 Strap withdrawing induction
I06 Strap feeding induction
I07 Strap pre-feeding induction
P00
P01
P02
P03
P04
P05
P06
P07
PC
Rev
Approved Date
Remark$
DRAWN
DESIGNED
CHECKED
APPROVED
SCALE
P 04
TITLE
PC Input Control
DRAWING NO
MH-101A/101B Strapping Machine
ISSUED DATE


PC Output Control
Model
MH-101A/101B
Controller
PC
K7M-DR14UE
PC
P45
P44
P43
P42
COM2
P41
COM1
P40
COM0
AC
AC
Q03
Q02
39
Q01
39
Q00
39
31
0
KM2 Strap pre-feeding motor control
KM4 Strap feeding motor control
KM3 Strap withdrawing motor control
KA2 Main motor control
FR1
35
KA1
33
SB1_1
SA1
OFF
Strap feeding
Strap withdrawing
P.01
0
31
P.01
31
P.01
0
P.01
Rev
Approved Date
Remarks
DRAWN
DESIGNED
CHECKED
APPROVED
SCALE
P 05
TITLE
PC Output Control
DRAWING NO
MH-101A/101B Strapping Machine
ISSUED DATE